# Supplementary Information: Reconstructing founder haplotypes, pedigrees and recombination from low-coverage sequencing of experimental crosses

Hadi Khan

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This supplement gives the finer rules of every stage, with the default value of each named parameter. Codes in parentheses (for example EC05 or AJ17) identify the method unit in the software's methods documentation (`docs/methods/` in the repository), where each rule is stated in full together with the code that implements it. Default values are those of the generated parameter reference (`docs/reference/parameters.md`) for release 4.0.

## S1. Read evidence, quality control and relatedness

### Input admission and the sample axis

A record is admitted when it has one ALT allele and both alleles are single nucleotides (A, C, G, T). All admitted records sharing a position are removed before AD is decoded. Missing AD gives zero reads. Input FILTER is not a quality decision (EC01). The sample axis is sorted by name, with the same permutation applied to every per-sample array, and outputs are restored to input order (EC03).

### Read-model fitting

*Folds (EC17).* Per-sample (depth, ALT) histograms of cells with $1 \le d \le$ `maximum_fit_depth` $=$ 48 are kept separately for training cells (fold $\lfloor x_j / w_{\mathrm{f}} \rfloor \bmod 4 \in \{0, 1\}$, $w_{\mathrm{f}} =$ `fold_window_bp` $= 1$ Mb) and held-out cells.

*Shared binomial (EC18).* EM starts from $(e, q) \in$ `em_starts` $= \{(0.02, 0.5), (0.01, 0.35), (0.08, 0.65)\}$ with genotype weights 1/3; pooled M-steps set $e$ and $q$ to expected wrong-allele (heterozygote ALT) reads over expected homozygote (heterozygote) depth, requiring $0 < e < q < 1 - e$.

EM stops when the gain is at most `em_tolerance_per_observation` $= 10^{-10}$ per observation. A start fails at `em_maximum_iterations` $= 2{,}000$, or on a likelihood decrease of more than $10^{-7}$ per observation. Each endpoint is then Newton-polished to stationarity in at most 40 box-constrained steps.

Endpoints are equivalent within relative distances of $10^{-13}$ (shared logits) and $10^{-8}$ (nuisance logits). The representative endpoint has the largest training log-likelihood; ties go to the lexicographically smallest parameters. A distinct endpoint within 1 nat of the best makes the fit unresolved. A failed polish fails its start; an unresolved fit, or failure of every start, gives the fixed model.

*Pooled sample binomial (EC19).* Each sample is first fitted without a prior, by L-BFGS-B (Byrd et al. 1995) plus a Newton polish. The variances $v_i$ of its two logits are diagonal entries of the inverse observed Hessian, so its genotype mixture is profiled out. They are used only if the smallest Hessian eigenvalue exceeds $10^{-8}$. For each logit $y$, $(\mu, \tau)$ minimise

$$\tfrac{1}{2}\sum_i [\log(v_i + \tau) + \frac{(y_i - \hat{\mu}(\tau))^2}{v_i + \tau}], \qquad 0 \le \tau \le \max\{1, 4\,\mathrm{var}(y)\},$$

where $\hat{\mu}(\tau)$ is the precision-weighted mean. The minimiser is refined to a root of the derivative. $\tau = 0$ is kept when its profile value lies within a relative $10^{-10}$ of the optimum.

With fewer than three finite variances, $\mu$ takes the shared value and $\tau = 0$. Samples are refitted under $\mathcal{N}(\mu, \tau)$; a $\tau = 0$ effect is fixed at $\mu$, and a sample without training reads takes the prior mode.

*Joint beta-binomial (EC20).* The fit starts from the pooled effects with $\kappa =$ `initial_heterozygote_concentration` $= 40$, under the pooled prior. L-BFGS-B is followed by a Newton polish on the arrowhead Hessian: one 4×4 block per sample, coupled to $\log\kappa$. The bounds are $\mathrm{logit}\, e \in [-16, -10^{-8}]$, $\mathrm{logit}\, b \in [-12, 12]$, $\log\kappa \in [-8, 16]$ and mixing logits in $[-25, 25]$.

*Selection (EC21–EC22).* Stretches without admitted cells are dropped. Complex candidates with non-finite held-out totals are skipped, and a failed complex fit never discards a valid simpler one. Acceptance needs $n_h \ge$ `selection_minimum_blocks` $= 4$, $\Delta > 0$ and $z \ge$ `selection_z` $= 2$ (inclusive), and a rejection does not stop later comparisons. The fallback model is binomial, with `fixed_error` $= 0.02$ and balance 1/2.

*Application (EC16, EC23).* Emissions are exact integer-depth rising products, exact in the binomial limit $1/\kappa = 0$. The `lookup_depth` $= 64$ table changes cost only. Validation requires exactly 1/3 at zero depth and finite, nonnegative values normalised within $10^{-12}$ elsewhere.

## Site filter

*Order (EC11).* All sites are kept, with the reason recorded, when the filter is off (the default for simulations) or when there are fewer than `minimum_calibration_sites` = 1,000 SNPs. Otherwise r1, r2, r3, r4, an initial r5, the library check and the r5 recount run in that order. Removal is the logical OR of r1–r5, or of r1–r4 when `drop_uninformative_sites` = false; the default is true.

*Classification (EC04).* The classification uses `classification_error` = 0.01 and `classification_heterozygote_alt` = 0.5. Frequencies start at 1/3 with a floor of $10^{-6}$. EM runs for at most 60 iterations, stopping from the fifth on once the gain falls below $10^{-7}$. $W_j$, $D_j^{\mathrm{hom}}$, $A_j$ and $H_j$ are moments from the last E-step. The leakage terms use the final frequencies:

$$\lambda_j = \sum_{i:\, d_{ij} > 0} [f_{j1} \sum_a \mathrm{Bin}(a; d_{ij}, \tfrac{1}{2})\, w(a, d_{ij}) + f_{j0}\, w(0, d_{ij}) + f_{j2}\, w(d_{ij}, d_{ij})],$$

$$\gamma_j = \sum_{i:\, d_{ij} > 0} d_{ij} [f_{j0}(w(1, d_{ij}) - w(0, d_{ij})) + f_{j2}(w(d_{ij} - 1, d_{ij}) - w(d_{ij}, d_{ij}))],$$

where $w(a, d)$ is the posterior-weighted wrong-read count of a cell.

*Clean core (EC05).* The histogram covers the 0.5th–99.5th percentiles with bin width $2\,\mathrm{IQR}\,n^{-1/3}$ (Freedman and Diaconis 1981). Integer data use rounded widths of at least 1; continuous data use 50–2,000 bins. A moving average over `histogram_smoothing_bins` = 5 bins smooths the counts. The mode is the first maximal bin, a reversible convention; tied modes are counted. Half-maximum distances divided by $\sqrt{2\ln 2}$ give the two standard deviations.

The local posterior is $\min(1, \hat{f}_0/\hat{f})$. Here $\hat{f}_0$ is the side-specific Gaussian at the peak density, and $\hat{f}$ is the smoothed density normalised by all finite observations, including those outside the histogram range. The window is the contiguous run of bins around the mode with posterior at least `core_posterior_level` = 0.5. It is unbounded where it reaches a histogram edge.

*r1, r2 (EC06–EC07).* r1 rejects only depths strictly beyond the window. r2 sums depths over the r1-passing sites before dividing, and fits its core to $\log X_j$ at the r1-passing sites with $X_j > 0$. Without such sites, r2 is not applied.

*r3 (EC08).* The domain is sites passing r1–r2 with $D_j^{\mathrm{hom}} \geq 1$ and $\gamma_j > 10^{-9} D_j^{\mathrm{hom}}$. The fit starts from clean weight 0.6, $\varepsilon_0 = \mathrm{clip}(\frac{1}{2}\,\mathrm{median}\,[\max(W_j - \lambda_j, 0)/\gamma_j], 10^{-5}, 0.1)$, $\varepsilon_1 = 10\varepsilon_0$ and $\phi = 1$. Each iteration updates three blocks:

- the weight, held within $[10^{-6}, 1 - 10^{-6}]$;
- $\varepsilon_0 \in [10^{-12}, 0.5]$, by safeguarded Newton steps on its concave objective;
- $(\log \varepsilon_1, \log \phi)$, by projected Newton ascent within $\varepsilon_1 \leq 10$ and $\phi \in [10^{-3}, 10^{4}]$.

It stops when the summed relative change falls below $10^{-5}$, or after 300 iterations. The rule applies if $\varepsilon_1 \geq$ `minimum_artefact_separation` $\times \varepsilon_0 = 3\varepsilon_0$. It removes sites whose clean posterior is below `clean_posterior_level` = 0.5.

*r4 (EC09).* The domain is sites passing r1–r3 with $H_j \geq 1$. A component has mean $m$ and inverse concentration $t$, with latent SD $\sqrt{m(1-m)t/(1+t)}$; $t = 0$ is binomial. Five fixed starts around the pooled $\sum A_j / \sum H_j$ are each fitted by L-BFGS-B, with at most three restarts and eight Newton polish steps. A start is certified when its projected gradient is at most $10^{-8}$; the best start has the lowest objective among certified starts (among all, if none is), ties by parameters, then start. The clean component has weight $\geq 1/2$.

With scale $\sigma_{\mathrm{sep}} = \max\{\text{clean latent SD}, \sqrt{m_0(1-m_0)/\,\mathrm{median}\,H}\}$, the rule applies when the artefact latent SD, or $|m_1 - m_0|$, is at least $3\sigma_{\mathrm{sep}}$. It abstains when $|m_0 - \frac{1}{2}| - |m_1 - \frac{1}{2}| \geq 3\sigma_{\mathrm{sep}}$, or when no start is certified. This contig-level gate can miss well-resolved high-depth sites.

*r5 (EC10).* r5 rejects a site with fewer than `minimum_expected_carriers` = 2 expected carriers (a strict inequality). After library flags, it is recomputed on unflagged samples only. r1–r4 are not recalibrated.

## Library quality check (EC12–EC15, EC25)

The check runs only where the site filter applies (by default, on for VCF input), on the sites passing r1–r4.

*Eligible cells.* The cohort-minor allele is ALT when cohort ALT reads do not exceed REF reads. A library carries it at a cell with $\tilde{a}$ minor reads of depth $d$ when $\Pr(Y \geq \tilde{a}) \leq \eta/(N-1)$, where $Y \sim \mathrm{Bin}(d, \varepsilon_0)$ and $\eta$ = `carrier_tail_level` = 0.05. Here $\varepsilon_0$ is r3's clean rate, or the classification error when r3 did not fit. A cell is eligible when the library is observed there and no other library carries the minor allele; cells with $\tilde{a} = 0$ count.

*Weights and partitions.* Because nearby cells share reads, each eligible cell is weighted by the reciprocal of the library's eligible-cell count in its `fragment_bp` = 1,000 bp window. Sites are cut into `artefact_regions` = 16 regions of equal site count, and each library's cells into `heterozygote_depth_classes` = 4 depth-quantile classes.

*Mixture components.* The null mixture has, per depth class:

- binomial error components on a log grid from $0.1\varepsilon_0$ to $\alpha_{\min}$, with ratio 1.6;
- beta-binomial own-heterozygote components in both orientations, with means at r4's clean balance (or at the ends and midpoint of r4's two means when its labelling is unresolved) and latent dispersions {0, 0.0001, 0.0004, 0.0016, 0.0064, 0.0256, 0.04};
- per-region artefact shares, under which minor counts are uniform on $0..d$.

The alternative adds $c_\mathrm{f}\,\mathrm{Bin}(\tilde{a}; d, \alpha)$, with share $c_\mathrm{f}$ common to depth classes and regions. $\alpha$ takes 24 log-spaced values in $(\alpha_{\min}, \alpha_{\max}]$, where $\alpha_{\min}$ = `minimum_foreign_fraction`/2 = 0.005 and $\alpha_{\max}$ = `heterozygote_floor_ratio` $\times\, h_{\mathrm{lo}}/2$, with $h_{\mathrm{lo}}$ the lower end of the own-heterozygote balance interval. The mixtures are fitted by EM with at most 400 iterations and tolerance $10^{-10}$. Alternatives are warm-started in ascending $\alpha$, and the null is re-polished from the best alternative's weights.

*Flagging.* A library is flagged when $\Lambda = \max_\alpha 2[\,\ell^{(1)}(\alpha) - \ell^{(0)}\,] \geq u_N$, the difference of maximised alternative and null log-likelihoods at the first maximum. The threshold $u_N$ satisfies $\Pr(\chi^2_1 \geq u_N) = 2\eta/(24N)$. This is a Bonferroni level under the boundary law $(\chi^2_0 + \chi^2_1)/2$ (Self and Liang 1987). It is nominal only for a regular, correctly specified likelihood and is documented as a working threshold.

*Inapplicability.* The check is recorded as inapplicable when any of the following holds:

- fewer than `minimum_samples` = 20 samples have observed cells;
- no cells are eligible;
- $2\varepsilon_0 \geq \alpha_{\min}$;
- $\alpha_{\max} \leq 1.25\,\alpha_{\min}$.

*Effect of a flag.* A flag hides the library's cells only in the founder-construction view and in r5.

*Inherited segments (EC15).* A diagnostic report, never a flag or parent call, groups near-private cells shared with one partner (gaps below `segment_gap_bp` = 1 Mb; at least `segment_minimum_events` = 3 events).

*Discovery sites and blocks (EC24).* Discovery sites have an integer total depth of at least `minimum_total_depth` = 5. Blocks are runs of `block_sites` = 200 consecutive retained sites, the last possibly shorter, and ignore spacing and the discovery mask.

## Genetic map (EC27)

A map needs at least two knots, with finite, nonnegative, strictly increasing positions and finite, nondecreasing cM values. The model coordinate interpolates between knots and extends beyond them at `rate_cm_per_mb` (default 5.0). Without a map, it is that rate times position. The measured coordinate equals the model coordinate if `rate_is_measured` is true. Otherwise (the default), measured coordinates clip positions to the terminal knots, and a chromosome without a map has zero measured length. The Haldane fraction is computed as $-\frac{1}{2}\,\mathrm{expm1}\,(-2\delta)$ (ledger entry L1); each consuming model sets its own probability bounds. The TroMau configuration supplies no map file and uses the 5 cM/Mb fallback with `rate_is_measured` false.

## Tie rules and shared emissions (EC30–EC33, EC35)

Shared tie quanta are powers of two, set per decision kind from the decision's scale and the contrast's measured motion under $10^{-6}$ relative genotype-likelihood noise. Classes link consecutive sorted exact values. A class wider than `WIDE_CLASS_QUANTA` = 4 quanta is ordered by value inside. Improvement is strict; content keys are exact discrete data. Reversal flips only arbitrary conventions, such as the histogram-mode tie, and enters the derivation. The read-model representative and r4's best start use lexicographic parameter order. Relatedness uses `gain_tolerance` = $10^{-6}$ nats with evidence keys.

Founder-pair emissions (EC35; assembly and founder refinement) score a sample's normalised triple $P$ under an ordered founder pair as $P(g)$ for called dosage $g$, $\frac{1}{2}[P(0)+P(1)]$ or $\frac{1}{2}[P(1)+P(2)]$ for one called allele (0 or 1), $\frac{1}{4}P(0)+\frac{1}{2}P(1)+\frac{1}{4}P(2)$ for two unknown alleles of different rows, and $\frac{1}{2}[P(0)+P(2)]$ for two copies of one unknown allele (a per-observation approximation). The robust score of a predictive likelihood $\rho$ is $\max(\log(0.99\rho+0.01/3), -2) - \log\frac{1}{3}$; uniform or zero-sum triples contribute nothing.

## Relatedness (RL01–RL19)

*Inputs (RL01–RL03, RL05).* Contigs are concatenated in configuration order, fixing ranks and reduction order. Allele frequencies take `iterations` = 6 EM steps from `initial` = 0.5, with update $p \leftarrow N^{-1}\sum_i[\frac{1}{2}\omega_{i1}+\omega_{i2}]$ ($\omega_{ig}$ posterior genotype probabilities) clipped at `clip` = $10^{-4}$. Every sample counts in $N$, and unobserved samples contribute their prior weights. Site gates are `minor_allele_frequency` = 0.05 and `minimum_observed` = 0.5, both inclusive. Thinning keeps every `family_site_step` = 20th passing site, starting from the first. Log terms are floored at $10^{-300}$.

*Keys (RL07).* A sample's evidence rank is the dense lexicographic rank of its full child-term vector, ordered by site and then by genotype pair. Equal vectors are exchangeable. A family key lists member ranks, then parent ranks (latent = −1), each in descending order. A pair key is the two family keys, sorted. Ties within `gain_tolerance` go to the larger key.

*Agglomeration (RL09).* Pairs whose full-sib gain is strictly positive and strictly above their half-sib gain are merged best-first, with affected entries rescored, until none remain.

*Parent assignment (RL10).* Each family's first free slot is offered the best sequenced non-member (excluding members and the other parent). In families of two or more, each member $i$ is also offered: $i$ moves to its best other placement and fills the slot. Offers are ranked on the sweep's starting partition by gain, parent rank and family key, and applied only if:

- its rescored gain exceeds $\log N$;
- neither its family nor its parent has been used in this sweep;
- its slot is still free;
- the parent graph stays acyclic.

No second-best search follows a failure.

*Moves (RL11–RL12).* A sample may not join a family whose sequenced parent is itself or its descendant. The best-improvement search applies, up to $N$ times per sweep, the rescored largest move gain above $10^{-6}$ (ties by evidence rank, then target key, the singleton target included). The content-order search visits each sample once, by decreasing rank, moving it when its best gain exceeds $10^{-6}$.

*Merges and sweeps (RL13–RL14).* Two families are compatible when their sequenced parents number at most two, none being a member of either family. Latent pairs must also beat their half-sib link. Offers above $10^{-6}$ are ranked by gain, then pair key. The first acyclic merge is applied, and a cyclic trial is undone without reordering. A sweep consists of parent assignment, moves and one merge, up to `max_sweeps` = 30. Convergence and the margin between the searches are recorded.

*Screens (RL15).* Float32 screens bound their error by $0.1 + 10^{-6}M$ nats plus $2(M+16)2^{-53}$ times the summed log-likelihood magnitudes. Comparisons the bound leaves undecided are evaluated exactly. Caches hold only gains of unchanged families; screens never change decisions.

*Support (RL16–RL17).* Each chromosome's measured length between family sites is cut into $\max(1, \lfloor \text{length}/1\,\text{M} \rfloor)$ equal units (`support_unit_morgans` = 1). Cuts map to the first site at or above them; empty units are dropped. A member wins a unit when its stay value, $\ell_F - \ell_{F\setminus i}$ (or $\ell_F$ if alone), beats both alternatives by more than $10^{-6}$. The alternatives are its best join gain $\ell_{G\cup i} - \ell_G$, excluding its own family and families it parents, and being alone with latent parents. A latent one-member family therefore scores 0. Parent support compares $\ell_F$ with the named parent and with a latent parent; it is missing for latent slots.

*Links and numbering (RL18–RL19).* After the kept fit, even one stopped at the sweep cap, every pair of fully latent families with a link gain above $\log N$ is reported, with no full-sib comparison required. Families are numbered by the rank of their smallest member.

*Pairwise IBD (RL04).* The chain uses every `ibd_site_step` = 10th retained site, without the family-site gates. For two samples' triples $P, P'$, $\psi = \psi_j$ and $p = p_j$, the state emissions are

$$o_0 = \Big(\sum_g \psi(g)P(g)\Big)\Big(\sum_g \psi(g)P'(g)\Big), \qquad o_2 = \sum_g \psi(g)P(g)P'(g),$$

$$o_1 = (1-p)^3 P_0 P'_0 + p^3 P_2 P'_2 + p(1-p)P_1 P'_1 + p(1-p)^2(P_0 P'_1 + P_1 P'_0) + p^2(1-p)(P_2 P'_1 + P_1 P'_2).$$

Sites not jointly observed emit 1. The chain starts uniform and stays with probability $\exp(-\theta\delta_j)$ over the model-map distance $\delta_j$ from the previous chain site, with $\theta$ = `ibd_switch_rate` = 4 per Morgan; it leaves to each other state with probability $[1-\exp(-\theta\delta_j)]/2$. Posteriors are averaged over the jointly observed sites. The IBD0 (IBD2) ratio compares the full chain with one lacking IBD0 (IBD2).

## S2. Local founder discovery and panel selection

### Local founder discovery

**Inputs (DI01).** Blocks hold `block_sites` = 200 sites; discovery sites have total depth of at least `minimum_total_depth` = 5 reads. Samples are sorted by name for the search and returned in input order. A block is informative when some discovery site is observed in some sample. The ceiling $\min(2N_a, 2^M)$ limits births only; there is no other cap on $K$.

**Tie quanta (DI02).** Quanta are fixed once per block from its null NLL $\mathrm{NLL}_0 = \sum_i \sum_{j\,\mathrm{obs}} [-\log \max_g p_{ij}(g) + 2\lambda]$, an upper bound on any panel's NLL. Each is the smallest power of two at least $\max(\mathrm{floor}, \mathrm{factor} \times \mathrm{NLL}_0)$: NLL comparisons $2^{-10}$ and $2^{-20}$ ($q_{\mathrm{nll}}$); A-step bands $2^{-13}$ and $2^{-26}$ ($q_{\mathrm{sample}}$); H-step allele changes $2^{-12}$; improvement tests $2^{-10}$ ($q_{\mathrm{imp}}$); release cells $2^{-14}$ ($q_{\mathrm{cell}}$); probabilities $2^{-15}$. Each floor is at least 16 times the measured motion of its quantity under a $10^{-6}$ relative perturbation of the genotype likelihoods, so no decision turns on numerical noise. Tie classes are formed by consecutive-gap single linkage; inside a class, stated preferences (such as fewer founders) precede an arbitrary content order, except that value order comes first in a class wider than four quanta. Budgets admit whole classes: a class straddling a budget is admitted up to twice the budget and omitted whole beyond it (recorded); classes ranked below a filled budget are ordinary truncations.

**Fixed-$K$ fit (DI03–DI09).** $\lambda$ = `unknown_copy_penalty` = 0.5 nats; $\tau$ = `switch_penalty` = 10 nats; `snps_per_bin` = 10 (one site per bin in blocks of at most ten discovery sites); likelihood floor $10^{-12}$; `max_iterations` = 50. The A-step band is $[\min, \min + q_{\mathrm{sample}}]$; the category preference (row pairs, $(k,U)$, $(U,U)$) is never reversed; the meet is order-free, e.g. $\{(k,k'),(k,k'')\} \mapsto (k,U)$. Samples without observed sites take $(U,U)$. In the H step homozygous carriers count twice in usage and once in evidence; unused rows keep their alleles; a comparison that an ordinary floating-point sum cannot certify is re-summed exactly. The second usage-tie schedule runs only if the first meets a tie. A cycle returns its lowest-NLL state (ties keep first-visit order) and is never reported converged. Each start contributes its assignment-only solution and refitted endpoints; endpoints with duplicate rows are refitted recursively at lower $K$. Panels are canonical (rows in lexicographic order), deduplicated by content (a later fit replaces an earlier only if better by more than $q_{\mathrm{nll}}$) and ordered by NLL class, smaller $K$, then content. Panels differing only at alleles no assigned carrier observes are one archive entry. Memoisation and fixed-point certificates (DI07) change work, not endpoints.

**Cavity score (DI10).** $a$ = `state_concentration` = 0.5, $\omega$ = `unknown_prior_mass` = 0.02, $\delta$ = `double_unknown_fraction` = 0.25; base masses are $(1-\omega)/K^2$ for $(k,k)$, $2(1-\omega)/K^2$ for $(k,k')$, $\omega(1-\delta)/K$ for $(k,U)$ and $\omega\delta$ for $(U,U)$, renormalised. A held-out sample tied among states $\mathcal{T}_i$ removes $1/|\mathcal{T}_i|$ from each. Mean-field ALT marginals $\mu_{kj}$ start from $h_{kj}$, $1-h_{kj}$ and 1/2; rows without remaining support stay at 1/2; a logit within $q_{\text{nll}}$ of zero is set to zero, keeping symmetric evidence symmetric; each start runs at most `mean_field_max_iterations` = 100 sweeps to `mean_field_tolerance` = $10^{-10}$. Starts are compared by training evidence lower bound within $q_{\text{nll}}$; all converged winners (all starts if none converged) form connected classes at $10^{-10}$; their predictive likelihoods, never their marginals, are averaged with equal class weights before the `likelihood_floor` = $10^{-300}$ and logarithm. No held-out evidence sets these weights. Unconverged solves are counted and block certification but remove no candidate.

$K = 1$ **starts (DI11–DI13).** Similarity is $\bar{s}_{ii'} = M^{-1}\sum_j \sum_g p_{ij}(g)p_{i'j}(g)$, distance $\max \bar{s} - \bar{s}$. Level-wise HDBSCAN* merges equal edges simultaneously and selects by excess of mass (`seed_min_cluster_size` = 2). A cluster's mean ALT fraction $\frac{1}{2}p(1)+p(2)$ is hardened at 1/2; within $2^{-15}$ of 1/2 the population ALT fraction decides. Seeds enter by whole cluster-size classes, largest first, up to `data_seed_count` = 8 (straddle 16); without cluster seeds, dosage rows of the most decisive samples (most observed sites, then largest summed maximum likelihood) are used. Fitted starts are kept by whole NLL classes up to `data_start_beam_width` = 6 (straddle 12); if every seed class is omitted the block is unresolved. Above `seed_landmark_budget` = 256 samples, clustering uses at most 256 landmarks lifted back to all samples (DI12); fitting and scoring still use every sample.

**Ordinary moves (DI14–DI16).** Deletions precede gauge rewires. Maximum cuts are exact per component up to `exact_cut_max_k` = 12 rows, otherwise found by degree-ordered local search (a recorded limitation); tied maximum cuts form one family with nominal budget 4× `max_cut_ties` = 16, admitted whole up to 32 and omitted whole above. Each cut yields a sitewise start (a site flips only if the samples with both copies on one side gain more than $q_{\text{nll}}$) and a start flipping all eligible sites. Deletions and rewires are admitted by whole families up to `max_proposals_per_expansion` = 128 (straddle 256). Data births rank samples, over both schedule endpoints of the parent, by classes ($q_{\text{sample}}$) of excess cost over the per-site genotype oracle, then more observed sites. Each sample supplies, for both extremal dosage families, subtraction seeds $\max(0, \min(1, \text{dosage} - h_k))$ for every row $k$ and direct dosage rows (dosage 2, or dosage 1 or missing where the population ALT fraction exceeds 1/2, becomes ALT). Whole tie groups are collected up to $\max(16, 2(K+1))\times$ `data_birth_attempts_factor` (3) starts; a group crossing `data_birth_tie_budget_factor` = 2 times that is omitted whole. Fitting stops at $(K+1)\times$ `data_birth_children_factor` (3) distinct complete children; collapsed endpoints are discarded; with no complete child, assignment-only fits are kept by NLL classes up to $K+1$.

**Soft-residual births (DI17–DI20).** Rows with at least `min_soft_responsibility` = 0.25 expected copies in a sample (diplotype posterior with a uniform pair prior) are subtracted; the residual ALT probability, given subtracted allele $h$, is $P(\cdot \mid h+1)/[P(\cdot \mid h)+P(\cdot \mid h+1)]$, called at `residual_call_probability` = 0.8. Distances are

Hamming fractions over jointly called sites (distance 1 below `min_residual_joint_known_fraction` = 0.10). Groups of diameter at most `max_cluster_hamming` = 0.10 are all HDBSCAN* tree clusters within the bound plus a greedy edge-clique cover of the remaining records, budgeted per connected component, `max_alternative_groups` = 64 (straddle 128). Pools are global and per (subtracted row, dominant partner) route; a record joins every route whose partner mass is within the probability band of its largest, so no convention selects one route. A consensus pools weighted log odds, one vote per sample capped at weight 1, and is emitted with at least `min_unique_sample_support` = 2 samples, effective support `min_effective_sample_support` = 1.5, a fraction `min_candidate_known_fraction` = 0.8 of sites called at `candidate_call_probability` = 0.9, and when it is no near duplicate (`dedup_hamming_fraction` = 0.005 over `min_dedup_joint_known_fraction` = 0.6 joint sites, coverage `min_dedup_bidirectional_coverage` = 0.95 both ways) of a panel row or better-supported emission. Candidates enter by whole support classes up to 128 (straddle 256) and are hardened at 1/2 as for $K = 1$ seeds. Replacements delete any row of a birth, taking births by NLL classes up to `max_replacement_children` = 48 starts (straddle 96).

**Archive, expansion and status (DI21–DI25).** Each $K$ keeps whole NLL classes up to `beam_width` = 6 (straddle 12); a wider leading class empties the beam while its last scored class stays selectable. Every fitted panel is remembered: one that has left the beam re-enters only with an NLL better by more than $q_{\mathrm{nll}}$. Refresh visits $K$ by $|K - K_{\mathrm{sel}}|$; new scores count against $96 + 12K_{\max}$ (`exact_score_backstop_*`), and a class that cannot be scored whole leaves the previous one in place. Score classes chain across $K$; in a class wider than four quanta only the exact maximum survives; the smallest $K$ in the class is selected, so another $K$ can link a class but never wins. All parents of a class are expanded before reselection. Replacements run after a phase without improvement above $q_{\mathrm{imp}}$; bridges run in the residual phase when $K + 2$ is within the ceiling and $K + 1$ is scored. Expansions stop at $32 + 8K_{\max}$ (`expansion_backstop_*`). Stop reason, omissions, certification, ambiguity, score margin (selected score minus the best other $K$) and unresolved status are recorded and never steer the search.

**Release (DI26–DI27).** `min_call_probability` = 0.85, `min_supporters` = 2. A carrier contributes, and counts once, only when its cost difference exceeds $q_{\mathrm{cell}}$ in magnitude. Tied winners are matched by known content: a row absent from any alternative is withheld, and a call survives only where all matching rows agree, with the least confident probability and smallest support.

**Cost.** An A step costs $O(NK^2B)$; a trajectory iteration $O(NMK^2)$ (at most 50 iterations, two schedules); the search $O(NMK^4)$ under exemption 0c ($K^4 = 810,000$ at $K = 30$); a mean-field sweep $O(NMK^2)$; release $O(NKM)$. Budgets are scientific parameters, independent of computing resources.

## Local panel selection

**Model (LS01–LS02).** $u$ = `unknown_mass` = 0.01; $\gamma$ = `redraw_rate_per_morgan` = 3, a redraw intensity rather than a crossover rate. Recursions are scaled and samples are summed in ascending order, so results do not depend on threads.

**Objective (LS03–LS04).** Context founders without calls are dropped; founders with equal rows remain separate claimers. Among descriptions within a relative $10^{-9}$ of the shortest, the largest matching is taken. Edit distances count called sites only, so one changed call moves the code by at most one edit increment.

**Fit (LS05–LS07).** `tolerance` $= 10^{-6}$ nats. Rows whose summed gains lie within a relative $2^{-10}$ (`FLIP_TIE`) of the largest are compared by the exact objective after flipping; if different panels remain tied the fit stops. Updates stop at $20 + 4K + 2M$ (`update_backstop_*`); rows are returned in canonical order. The trajectory-best state is the earliest strictly best; its frequencies are then refitted while the context objective rises. A snap proposes each called site where a matched row differs from its founder, unless the flip would duplicate a row, with bound equal to the exact one-bit gain plus the edit saving $\log V(n, d) - \log V(n, d-1)$. It needs a bound above $\eta$; bounds within $\eta$ of the largest form one class, decided by the smallest canonical panel; a snap is kept only if the objective rises.

**Search (LS08–LS15).** Deletions are proposed in row order, merges in lexicographic pair order and additions in bank order (new row frequency $1/(K+1)$); a single-row panel has no deletions or merges. A merge's consensus row takes the smaller row's allele within $2^{-20}$ of 1/2; the merged row carries $f_k + f_{k'}$, and a consensus equal to another row merges that row too. Starts with equal panels and frequencies within $10^{-12}$ are one start. The merge bound is the deletion panel's log-likelihood plus the kept row's expected gains at changed sites plus $\sum_k \bar{n}_k \log(f_k^{\mathrm{m}}/f_k^{\mathrm{d}})$ ($\bar{n}_k$ and $f^{\mathrm{d}}$ from the deletion panel, $f^{\mathrm{m}}$ the merged frequencies on its rows), the larger of two such bounds. Exactly scored merges are, per row, the best-bounded merge with a nearest row and the best-bounded merge not involving it whose consensus is nearest it. Each kind refits whole classes until `refits_per_kind` $= 8$; a class crossing `refits_per_kind_cap` $= 16$ is excluded. Refits reaching one panel are one candidate. Rounds stop at $10 + 2K$ (`round_backstop_*`). If a start's fit reaches the previous round's panel, the previous fit is kept unless beaten by more than $\eta$; if the previous fit is better by more than $\eta$, the search starts from it. A whole search is reused only for identical inputs; fits, scores and bounds are memoised across the first two rounds (LS13, changing work only). Outcomes of several starts are compared by objective class, then fewer founders; distinct tied panels become alternatives (LS14). A retained incumbent discards the attempted alternatives (LS15).

**Bank (LS16–LS20).** The initial round has no incumbent and no bank search. A proposal row with no call is dropped; a partial row takes, at uncalled sites, the alleles of every latent row at minimum Hamming distance; tied completions give one source per combination up to `completion_variant_cap` $= 8$, above which the source keeps only uniquely completed rows. A source's best set is the nested class of cavity score, then NLL, then fewest founders; each member is released by discovery's rule, with uncalled cells taken from the source's own latent panel. The bank search first fits the whole bank as one start, then visits deletions, additions and row replacements (closest first) with $\kappa =$ `complexity_scale` $= 0.5$, reselecting by nested classes of penalised score, NLL and founder count, and stopping without improvement above $q_{\mathrm{imp}}$ or after $8 + \lceil K/2 \rceil$ rounds. It fits at most $512 + 64K$ novel starts per call and admits at most $1024 + 128K$ panels, overlapping fit families entering by whole NLL classes (LS20). The search bank is the latent rows, starts, selected panels' rows and the previous panel's rows.

**Release (LS21–LS24).** `min_call_probability` = 0.85, `min_supporters` = 2, `favourable_margin` = $2^{-14}$ nats. Carrier probability counts a homozygous state once; a carrier whose own gain lies within the margin of zero is not favourable. A previous release is reused only for identical panel and frequencies without context witnesses. A site's unknown-copy count is the smallest count whose mass is within a relative `unknown_count_band` = $2^{-15}$ of the largest; samples without depth report 2; tied alternatives take the maximum. Rows shared by all alternatives keep agreed calls, mean ALT probability and smallest support.

**Context refit (LS25–LS28).** `context_levels` = (1, 2). An uncovered block receives its current calls without context founders, hence no witnesses, claims, trajectory-best adoption or snaps. A component must match its blocks' concatenated positions exactly; duplicate component rows give a consensus template. Bins are blocks, spaced by the map interval between block centres; founders have prior mass $(1-u)/K$ and the background state $u$. Uncalled founders and the background carry the site's ALT frequency $(1+n_{\mathrm{ALT}})/(2+n_{\mathrm{called}})$ over called founders; the emission mixes the expected likelihood with 1/3 at weight `emission_mix` = 0.01. A site's founder-allele posterior, under a uniform prior, is enumerated exactly for at most 10 founders; above that, a belief-propagation mode and local ascent fix certain founders (flips costing over 20 nats), withhold multimodal sites, and enumerate the remaining core exactly (at most 10 founders) or within a Hamming ball of at most 1,024 configurations (radius at least 3), with a shell guard. Calls are released at `posterior_threshold` = 0.99 with count at least `min_support` = 2 or mass above $2+2^{-20}$; other cells keep the component's call, then the template. Duplicate local rows sum support and record multiplicity. Recorded summaries do not affect decisions (LS29–LS31).

**Cost.** A forward–backward pass is $O(NMK^2)$; a fit at its backstop $O(NMK^2(K+M))$; a climb $O(NMK^3(K+M))$ under exemption 0c; context-coded screening $O(K^5)$ per round under exemption 0 ($K^2$ merges each with an $O(K^3)$ matching; $24,300,000$ at $K=30$); snaps $O((K+M)^2)$ passes; context refit $O(NBS^2)$ time and memory per component.

## S3. Assembly, founder release and refinement

Parenthesised identifiers are the software's method units; defaults are those of release 4.0 (none is a user setting). Notation: $T$ bins of a batch; $W$ units of a proposal window; $b_w$ surrogate bins of a window; $R$ the window's row-alphabet size.

### Founder completion

**Folds (U03).** `fold_seed` (1729) is hashed with each sample's observation-mask content and depth key; equal-key groups, visited in byte order, go whole to the lighter fold (ties and inactive groups to fold zero). No random draw or label chooses a fold.

**Cross-fitted variational EM (U04).** Samples copy unordered row pairs and unresolved alleles are binary variables shared across samples, both under uniform priors; emissions are $\max\{\log(0.99\bar{p}+0.01/3), -2\}$. The variational family factorises over samples (pair states) and sites (configurations). A site's support is the full cube up to `MAX_EXACT_BITS` (10) unresolved founders. Above that it is an uncertain core fixed from the first iteration,

so the ELBO stays a lower bound. One start runs 2–100 iterations (ELBO tolerances $10^{-8}$; a decrease beyond $10^{-8}$ is an error). Training-fold calls need posterior ≥0.99, ≥2 carriers, non-exchangeable founders, a unimodal site and a passing shell guard. The held-out fold is painted only where the released panel calls every founder; absent training data leaves the pre-fill panel (recorded).

**Cavity fills (U05).** Bins hold 50 SNPs (`snps_per_bin`), and outside-bin statistics are prefix plus suffix sums. A founder is ineligible if another founder has an identical outside-bin allele vector (tolerance $10^{-12}$) or differs from some founder at fewer than one observable outside-bin site. Support, over focal-informative samples sensitive to the target, is the larger of the number with leave-bin-out carrier probability above 1/2 and the sum of those probabilities. A fill needs support ≥ 2 and posterior ≥ 0.99. A single-bin multi-founder block therefore gets no fill.

**Site posteriors (U08–U10).** Dosage polynomials are enumerated exactly; a value cancelling below $10^{-10}$ is recomputed as a direct positive sum. Above ten unknown founders, damped loopy belief propagation (damping 0.5, tolerance $10^{-9}$, ≤500 iterations) (Murphy, Weiss, and Jordan 1999) is followed by deterministic ascent from its mode and complement (backstops 50), on an uncertain-core cube or a Hamming ball (≤1,024 configurations, radius ≥3, 3-nat shell guard). Non-converged BP is flagged; this can withhold a fill even when a mode exists.

**Boundary links (U06).** Rows with identical pre-fill vectors form classes. Two classes across a boundary are linked only if, on samples informative on both sides, the mean squared discrepancy of expected copies and of $\min(\text{copies}, 1)$ is ≤0.099; the pair is a reciprocal unique minimum by margins 0.03 (pooled) and 0.01 (each fold); there are ≥6 pooled and ≥3 per-fold informative samples; each class has ≥1.0 pooled and ≥0.5 per-fold effective copies and one or more anchor sites separating it from the other classes of its block; and both blocks have two or more classes. Exact best/runner-up ties are not broken by row number.

**Occupancy gate (U07).** A block releases nothing if selection left unexplained copy mass above `maximum_discovery_unknown_mass` (0), or if it has a single founder.

*Abundant route:* squared carrier probabilities sum to ≥8 and ≥5 samples carry the row with probability ≥0.8; all its candidates are released (a sum within the tie band of 8 is not abundant). *Rare-continuity route:* through an accepted single-row link, ≥2 samples informative on both sides carry the row (≥0.9) and the neighbour (≥0.8), product ≥0.72; the best neighbour must be unique (tied neighbours abstain only when their qualifying samples differ); fills are released only where a qualifying sample is focal-informative.

Released fills publish the allele with posterior above 0.5.

## Hierarchy

**Partition (U11).** Linking proxies stride to at most `max_sites` (2,000) usable sites per component. A segment of $n$ components is cut into $m = \lceil n/10 \rceil$ batches, plus one when $n$ is odd and $m$ even. Sizes are near-equal, with the larger ones in mirrored pairs. Breaks

persist. A batch with one component or no common informative sample passes through unchanged.

**Levels (U02, U12).** Stop reasons, checked on unrefined join output, are in order: single component; no reduction (level ≥2); only phase breaks (level ≥ `minimum_levels`, 4). Joined founders are sorted lexicographically, so later slot tie-breaks are content rules. Final-role refinement follows every level, with additions only at the level whose stop reason is set. First-level paths, refined first and then unrefined, form the coarse alphabets of later refinements. Local selection runs one or two unrefined levels as context.

**Alternatives between levels (U13).** A family is an input choice only if it covers a component's exact span and contains that component's nominal panel; otherwise it stays unresolved. If the product of choices is at most `BRANCH_LIMIT` (16), every joint branch is joined on the sites usable under all choices and one common cohort; each count's winners are its top LL tie class, and the representative has the smaller $K$, then fewest exact switches, then content order. A larger product is recorded as an unresolved joint budget; a tied family is never reduced to a convenient subset.

## Linker and beam

**Scans and EM (AJ05–AJ07).** Emissions are $0.99\,\Pr(\text{obs} \mid \text{class}) + 0.01/3$ over seven predictive pair classes, without a floor. Inside a component a homologue switches with $\theta$ equal to the interval's map distance in Morgans (bp times the fallback rate without a map), capped at 0.5; weights $(1-\theta)^2$ (stay) and $\theta(1-\theta)/(K-1)$ per other founder are normalised. An error-tract chain has stationary fraction 0.01 and mean tract 20 kb; error-state emissions are the mean of the three likelihoods. EM per gap starts uniform; the M-step adds 0.1 pseudocounts, floors logs at $-10$, renormalises and mixes 0.01 uniform; it stops at $\max |M(\mathbf{A}) - \mathbf{A}| \le 10^{-3}$ or after 50 full updates (recorded as unconverged).

**Horizon and beam (AJ08–AJ11).** The horizon is $G = \max\{1, 1 + \lfloor 0.5/(3\bar{D}) \rfloor\}$ (`recombination_tolerance`, `gap_horizon_scale`). The beam builds paths forward with 16 per endpoint row, then refines block by block backwards, and archives every selected path. With constant transitions (AJ10, retained accretion), only $O(BK)$ first-covering paths are proposed. Linkage (summed step scores) is consulted only within LL tie classes.

## Join panel search

**Objective (AJ01–AJ04).** The bin size is $\max\{\min(100, \max(10, \lfloor \bar{L}/20 \rfloor)), \lceil L_{\text{span}}/2000 \rceil\}$, with $\bar{L}$ the mean number of sites per input. The switch source is the first maximiser in pair order, and a state stays on equality. The exact minimum switch count is only a secondary prior. Proposals come from tracebacks under both conventions.

**Start and tier 1 (AJ12, AJ15, AJ16).** All anchor panels are scored. Proposal kinds run in order: replace, static replace, novel replace, local, one-sided, splice, cycle, switch-splice, switch-cycle. An exploratory painting at penalty $\min(\lambda, 10)$ exposes coordinated switches for proposals only. At most `full_scores_per_kind` (16) edits per kind are scored exactly (a cut class kept whole, up to twice that). A move improves when $2\Delta\text{LL} > \max\{10^{-8}, 10^{-9}(1 + 2|\text{LL}|)\}$.

**Admissibility (AJ14).** A move must keep $K$ distinct paths, and each changed path must keep a private cell and stay more than `near_identity` (0.01) from every other path. Admissibility is checked on the edit witness, before deduplication; a seed and its copy are exempt during their refit.

**Bounds and tier 2 (AJ17–AJ20).** Exclusion forwards, the maxima without up to two paths, are recomputed only where those paths cover a bin's maximiser set; this is exact under ties. Two-region bounds limit removals of up to two paths with additions of up to two. Only a certifying bound skips work; samples whose bound exceeds their base by $10^{-9}(1+|\cdot|)$ are scored exactly. Tier 2 enumerates pool replacements and the local, one-sided and splice moves in canonical order, scores them best-first and records flat neighbours.

**Pool, plateaus, replays (AJ26, AJ28, AJ29).** Quotas scale with row multiplicities. The quota ladder doubles to 256, and to 1,024 after an acceptance at 256. A pool with identical ordered content triggers no count work. Plateau continuation follows uncertified winning-family members. Up to `MAX_BRANCHES` (4) searches per join, the primary included, replay forced choice prefixes (FIFO); a mismatch or exhausted budget records incomplete resolution, and seed choices are never forced.

**Backstops (AJ25, retained accretion).** With $K_b$ the founders of input $b$, a search stops after $100 + 20 \max_b K_b$ sweeps. Births and seed swaps freeze after $50 + 10 \max_b K_b$ count changes, while deaths continue. Backstop stops are reported as uncertified; a final census of paths failing their own death test is reported only.

## Join count step

**Deaths and births (AJ21, AJ22).** Paths without novelty are deleted one by one, starting with the largest $\mathrm{LL}(\mathscr{P}f)$. Otherwise the best candidate is ranked by that value in wide tie classes, then by negative linkage, then by content, and is tested. A birth is scored only if it passes two necessary conditions on upper bounds $u_i \geq d_i \geq q_i$: the second-largest $u_i$ exceeds $c/(2\tilde{N})$, and $\sum u - \max u > c/(2\tilde{N})$. Survivors are scored exactly in decreasing bound order, until no bound can come within four quanta of the best accepted gain.

**Duplicate seeds (AJ23; exemption 0b).** Pure-mosaic pool paths that pass the bound screen indicate a chimera trap. When $K \leq$ `ranked_refits` (9), every path is a seed; otherwise the 27 strongest indicators by bound have their exact total gains tested, and the main coverers of passing indicators give up to nine seeds. Seeds are refitted concurrently by certified moves restricted to the seed and its copy, then settled (paths without novelty deleted, then a death test). A refit is accepted if it survives, is not smaller and raises LL (a seed swap if the size is unchanged).

**Cycles (AJ24).** Born-path tags follow same-count edits; deleting a tagged path freezes births, and is marked undecided if ambiguous in a cycle that began with a decided deletion. Revisiting a canonical count state freezes births and seed swaps.

**Bootstrap and shadows (AJ30–AJ33).** Bootstraps draw 2,000 resamples (level 0.95) from a stream keyed by content (decision kind and larger panel in joins; component end positions, size, round and kind in refinement), shared across a shadow family. Bands are

resolved only for reported decisions. A complete shadow family of up to 2,048 members is enumerated only if the total test passes. In joins, an unverified count remains a diagnostic.

## Proofs and bound tests

**Count bounds (AJ34, RC10).** Optimistic scores let each block choose a free panel, resetting between blocks. A block with $K+1$ rows takes its best one-row deletion. Deletion bounds need every founder to have a private cell. With $\underline{d}_i = F_i(\mathscr{P}_K) - U_i$ ($U_i$ the per-sample bound), a deletion round is skipped when $\sum \underline{d}/c - \frac{1}{2}$ and $(\sum \underline{d} - \max \underline{d})/c - (\tilde{N}-1)/(2\tilde{N})$ exceed the rounding tolerance and the replicate-wise percentile of $\sum \underline{d}/c - \frac{1}{2}$ is positive.

An addition round is skipped if the total bound gain $U - F$ is below $c/(2\tilde{N})$, or if every $u_i = U_i - F_i$ is (which also proves non-ambiguity). Skips are recorded as proven decisions.

**Shadow-envelope certificate (AJ35; retained accretion).** In refinement only, a proof is attempted for an over-budget family when the largest per-block product is ≤2,048 and local work ≤4 times ordinary work. Block messages maximised over local products give $U_i^{\sigma}$ bounding every shadow, so $q_i \geq F_i(\mathscr{P}^{+}) - U_i^{\sigma}$. The proof succeeds only if every combined field equals the incumbent's under every permutation of each bootstrap draw. A successful proof clears the count-unverified flag only.

## Refinement: objective and fixed-K search

**Workspace (RC01, RC02).** Objective sites are kept sites with a called prepared row; at an all-uncalled site, pairs would differ only through the allele prior. Samples without evidence contribute exactly zero and are excluded from $\tilde{N}$ ($F$ still sums all samples). Interval hazards are `reset_rate` (1 per Morgan) times map distance, floored at $-\log(1-10^{-15})$ and summed over skipped sites; the first informative site has no transition; costs are stored in single precision. A map is required; genotype error is fixed at 0.01. The recursion over unordered pairs is

$$\mu'\{a,b\} = \max(\mu\{a,b\},\ \max(\bar{\mu}_a, \bar{\mu}_b) - \kappa,\ \max\mu - 2\kappa) + \epsilon(a,b), \qquad \bar{\mu}_x = \max_y \mu\{x,y\}.$$

Exact ties prefer lower switch cost; a separate recursion counts switches (RS02).

**Surrogate and windows (RC03, RS04).** The bin size is $\max\{\text{floor}, \lceil L_c/2000 \rceil\}$. The floor is $\max\{200, \lceil L_{\text{chrom}}/2000 \rceil\}$ at levels 1–2 and 200 thereafter (`proposal_min_sites_per_bin`, `proposal_max_bins`). The Potts penalty is 20 nats. Fine windows of 8 blocks at stride 4 offer all prepared rows. Coarse windows of 4 first-level spans at stride 2 offer the distinct first-level paths.

**Window programme (RS05, RS06; exception 0e).** A box bound certifies already-optimal windows. Otherwise an exact frontier programme prunes by closed-message dominance ($10^{-6}$) and valid bounds. Past a frontier of 256 it restarts with Lagrangian suffix bounds (≤100 dual sweeps) (Rush and Collins 2012) and a feasibility beam of 16, keeping ≤512 children per unit; it is approximate where that cap binds, and such units are counted. A frontier beyond `frontier_bytes` (128 GiB) keeps the incumbent. The

canonical answer is the incumbent when it lies within $2^{-14} + 10^{-9}|H|$ of the optimum ($H$ the incumbent's surrogate value), and otherwise the lexicographically first path within that band.

**Work account (RS07, RC18; accretion, exception 0e).** Each template is allowed $2^{25} NWb_w^2 \max(K,R)^3$ charged units (`window_work_coefficient`). Exhaustion discards the template and stops the component, which keeps its input rows and families plus an unperformed-search family; during polish it keeps the unpolished panel. The bound is per template, not per component.

**Acceptance (RS08–RS10).** At each junction, exchanges come first, at most 4 (`EXCHANGE_ROUNDS`). Content moves follow, fully re-ranked through 4 acceptances (`CONTENT_FULL_ROUNDS`) and then rescored singly. The threshold is TOL = $2^{-9}$ nats. A coarse move with $\Delta F + \Delta(\text{switch cost}) < -10^{-6}$ is guarded: affected paths are traced with and without it, switch events are moved to the leftmost site of their evidence interval, paired excursions are excluded, and the move needs $\Delta F - C_{\text{shared}} > \text{TOL}$ with $C_{\text{shared}} = \sum_i \min(\text{saving}_i, \text{cost of shared one-way removed events})$. Fine moves and bridges bypass the guard.

**Fits (RS11, RS12).** A fit converges at a sweep that accepts nothing. Reaching `sweep_limit` (100) records nonconvergence. An $F$ that deviates from its prediction by more than $10^{-9} \max(1, |F|) + 10^{-6} \times$ (accepted moves) is an error. Of the two direction starts, the winner is the higher $F$ class, then the minimum switch count; tied variants are retained, and rows are canonicalised.

## Refinement: count pass

**Cleanup (RC14; retained accretion).** Before each deletion round, a founder is removed if its unrefitted deletion changes every informative score by at most $2^{-9}$; the panel is then refitted. Scores include $K$-dependent normalisers, so this rarely fires. It is not a general test of usage.

**Rounds and shortlists (RC09–RC11; exemption 0b).** Deletion candidates are the founders without a private cell if any exist, otherwise all founders. Above nine candidates, deletions are ranked by $\min(m, m')$ of their exact unrefitted deletion, and addition seeds by copy best-response gain (RS13: duplicate the founder, solve fine windows for the copy, run one forward unguarded content pass). The top nine are kept with the cut's tie class, which is dropped if it would exceed 18 members unless it is the first class. The best-$F$ refit is decided. Additions stop at twice the incoming $K$.

**Symmetric null and ties (RC12, RC13).** The death class is the top tie class of unrefitted $F(\mathscr{P}^+ \setminus f)$, taken over founders without a private cell if any exist. Every member is evaluated with one bootstrap stream; disagreement or an unresolved member family yields $\mathscr{P}^-$ with ambiguity. Every member of the selected refit's tie class is also decided. Opposed verdicts keep the smaller-panel outcomes, which are ranked by margin, then by fewest switches.

**Restoration, terminal family, continuation (RC15–RC17).** A fit is remembered at its $K$ only if it improves by more than $\max(\text{TOL}, \text{quantum})$. At the final $K$ a decisively better fit is restored and its adjacent-$K$ comparisons rerun. The returned panel's $F$ class is rescored exactly; its fewest-switch member represents it and the live family is released. Continuations replay alternative input roots and count choices breadth-first from the original input, within `alternative_branches` (16) leaves. Winning classes are compared per count, preferring the smallest $K$, then the fewest switches. Overflow stays unresolved.

**Polish (RC19, RS14–RS17).** Polish applies one-direction fits to the decided panel and to each distinct live alternative. At each ordinary fixed point it tries a beam escape (width 64); if nothing is accepted, a long repair (one-founder moves over 64-block windows at stride 16, beam 16, each with an optional 8-block private correction from the two most-agreeing donors or the incumbent); if still unchanged, a bridge (fine precursor plus suffix exchange). Accepted gains are verified exactly; counts are not re-evaluated.

## Ties

Each quantum is the smallest power of two that is at least $\max(\text{floor}, \text{factor} \times |\text{scale}|)$:

| Kind | Floor | Factor |
|---|---|---|
| `join_ll` | $2^{-19}$ | $2^{-34}$ |
| `join_ll_wide` (deaths) | $2^{-19}$ | $2^{-30}$ |
| `join_proxy` (bounds) | $2^{-19}$ | $2^{-24}$ |
| `beam` | $2^{-18}$ | $2^{-30}$ |
| `refine_F` | $2^{-9}$ | $2^{-26}$ |
| `count_margin` | $2^{-26}$ | 0 |

A class chains values whose consecutive gaps are at most one quantum, and a class wider than 4 quanta keeps value order (U14, AJ27, RS09). Within a class, linkage decides first (joins only), then content. Convention reversal changes only arbitrary content orders, never a prior.

## Founder release (U16)

Alleles that no painted carrier observes are withheld first. For each retained family, blocks whose row multisets differ are isolated and their unmatched rows withheld; greedy forward and reverse scans cut continuity wherever the panels stop agreeing up to a row permutation. Segments are reordered by content, and withheld rows are barred from the downstream r5 fill. Count-unverified records annotate segments without withholding. A family not matching the final panel is reported as unresolved correspondence.

## Cost class in K

| Mechanism | Dominant cost | Class |
|---|---|---|
| Completion | $O(NLK^2)$ cross-fit; $O(NK^3)$ ball sites; $2^{10}$ cube | $\le K^3$ |

| Mechanism | Dominant cost | Class |
|---|---|---|
| Linker | $O(NJK^2)$ scans ($J$ emission sites); $O(NK^3)$ propagation | $\le K^3$ |
| Join search | $O(\tilde{N}T^2K^3)$ per structural query; query count quartic in $K$ on a finite grid | exception 0e (AJ17) |
| Ranked refits and shadows | $\le 9$ refits; $\le 2{,}048$ shadow scores of $O(\tilde{N}TK^2)$ | exemption 0b |
| Refinement search | $O(NLK^2)$ messages; $O(BNK^3)$ exchanges; window fixing bounded by the work account | exception 0e (RS05, RS07) |
| Release | $O(Bn_{\text{alt}}K\log K)$, $n_{\text{alt}}$ alternatives | $\le K^3$ |

## S4. Painting, pedigree inference and founder refit

### Painting

Parameter names follow the generated parameter reference, values are 4.0 defaults, and method units are given in parentheses.

**Alphabet, bins and local keys.** Classes are exact call signatures, numbered by first appearance; unanchored classes and BACKGROUND share one public UNKNOWN identifier (PR01). Bins split the $L_s$ selected sites into $\max(1, \lfloor L_s/\texttt{snps_per_bin} \rfloor)$ runs, with $\texttt{snps_per_bin} = 100$; the first $L_s \bmod B$ runs are one site longer. Centres are mean bp positions, and edges are half-open, at site midpoints. Bins never depend on task windows or sample batches (PR02). In each bin, anchored classes called there with identical $\{-1, 0, 1\}$ vectors share their smallest index as a key; other anchored classes key to themselves, and the rest to UNKNOWN. Local equality does not imply chromosome-wide row identity (PR03).

**Emissions and transitions.** A site is informative when it is observed and its likelihood row spread exceeds $\texttt{INFORMATIVE_SPREAD} = 16\varepsilon_{\text{mach}}$. Sums run in site order and are divided once by the design effect. A bin split by label continuity keeps its whole-bin denominator in every part (PR04). $\texttt{MINIMUM_SWITCH_PROBABILITY} = 10^{-15}$. The switch probability $\theta_b$ never exceeds $(S-1)/S$, and edges of zero length are left out of the $\rho$ update. The rate $\rho$ is a pooled copying rate, not a count of meioses (PR05). Track states are the $S$ faithful states followed by one error state per anchored class, $S' \le 2K+1$. The diploid transition factorises over tracks, so recursions cost $O(S'^2)$ per bin and sample (PR06). Forward–backward is scaled at every step and reduced in sample order (PR07).

**Estimator.** Starting values are $\texttt{initial_switch_rate} = 2.0$, $\texttt{initial_error_entry} = 0.001$ and $\texttt{initial_error_exit} = 0.25$, with $\texttt{estimation_iterations} = 8$ updates (PR08). With $T$ eligible tracks and expected label changes $\bar{n}_b \le T$, the update for $\rho$ maximises

$$Q(\rho) = \sum_{b:\, d_b > 0} [\bar{n}_b \log \theta_b(\rho) + (T - \bar{n}_b) \log (1 - \theta_b(\rho))]$$

over RATE_BOUNDS = (0.01, 1000) per Morgan. The search takes RATE_GRID_POINTS = 241 log-spaced points, then GOLDEN_STEPS = 60 golden-section steps. The previous value is kept unless $Q$ improves; $Q$ can be multimodal, so the search is not a certificate. Let $s$ and $e$ be the expected error-run starts and ends, $k_f$ and $k_e$ the kept faithful and kept error transitions, and $\iota_e$ and $\iota_f$ the stationary draws into error and faithful states. The update for $(\alpha, \beta)$ maximises

$$(s + \iota_e) \log \alpha + k_f \log (1 - \alpha) + (e + \iota_f) \log \beta + k_e \log (1 - \beta) - (\iota_e + \iota_f) \log (\alpha + \beta)$$

over ENTRY_BOUNDS = $(10^{-6}, 0.5)$ and EXIT_BOUNDS = $(10^{-3}, 0.999)$. It uses L-BFGS-B in log space, from the previous values and from stay-only ratios, and keeps the previous values unless the objective improves. Results are clipped and rounded to a natural-log grid of estimate_grid = $2^{-6}$. Without an anchored class there is no error state, and $(\alpha, \beta)$ are reported at their bounds. A fitted $\alpha = 10^{-6}$ is a boundary fit, not a test of absence.

**Decoding.** Predecessors within VITERBI_TIE = $2^{-15}$ nats of the best are tied. Ties prefer staying in the same state, then the same label in its other fidelity state, then a label change from the smallest index (the largest under reversal); the final state is the smallest index within the quantum. The mirrored convention decodes the transposed chain (PR09). At ties between ordered and swapped continuations, the suffix is swapped if the swap gives strictly fewer allele-visible changes. A change between classes that share a local key is not allele-visible, and a pure swap is no switch (PR10). Switch intervals use a discrete test on calls, the observed mask and the shared-allele rule, and stop at neighbouring switches. The two conventions' masks are combined into the smaller unordered cover. Equal-size covers that differ give every candidate to both tracks. Canonical track order is then fixed from right to left: fewer changes, then fewer allele-visible changes, then the smaller mask on track 0. MAXIMUM_MASK_STATES = 63 requires $S \le 63$; at $K = 30$, $S \le 31$ (PR11).

**Release.** Only $q^F$ and $q^{AF}$ are computed, each the minimum over the two conventions (PR12). The threshold minimum_release_posterior = 0.9 is inclusive, with no tie band. Statuses take precedence in this order: ineligible, low posterior, switch interval, local equivalence, then the class status. Member sets are the class's rows (singleton or pooled), the rows of all key-sharing classes (local equivalence), or the union of the candidates' rows, extended to key-sharing classes in stage-2 bins and empty if any candidate is BACKGROUND or unanchored (switch interval). A switch interval cannot release a bin that failed the staged rule, and ineligible samples report zero switches (PR13); caching affects resources only (PR14).

**Label continuity.** Link points are the first selected sites of atomic spans. Only singleton anchored classes move, and the search needs two movable classes, a link and two bins (PR15). Exact transposition gains separate the unaffected states from the $4S - 4$ affected ones, using left and right max-product messages, and are summed in sample order (PR16). The unit $\Delta_{\text{sw}}$ is the median over edges of the log-probability of the ordered pair staying minus that of one specified single-track change. Candidates gaining at least label_continuity_minimum_switch_gain × $\Delta_{\text{sw}}$ = 2 units are ranked in descending tie classes. The tie quantum is the nll quantum: the smallest power of two at least

$\max(2^{-10}, 2^{-20} \times \text{largest gain})$ nats. A class wider than `WIDE_CLASS_QUANTA` = 4 quanta keeps value order; otherwise link and pair index decide. A placement moves left while the two rows carry identical founder-axis data between consecutive links.

Acceptance requires the painted-switch total to fall by at least 2, with losers at most `label_continuity_maximum_loser_ratio` = 0.1 times winners. A tie class is skipped when another acceptable member would relabel differently, and a candidate that changes a class's error capability is rejected. Every accepted move lowers the total, so the search terminates (PR17). Verification requires the relabelled panel to have identical classes, sites and anchoring and a strictly lower repainted total; otherwise the assembled panel is kept, with a recorded reason (PR18). The stage order is model, continuity, decode, release (PR19).

**Cost.** The chain costs $O(NBS'^2)$, masks at worst $O(NBS^3)$, and the screen $O(\text{links}\, NS^3)$ per iteration. Label-continuity checks have an $O(K^4)$ worst case, carried as documented exception 0e for review in release 4.1 (PR17).

## Pedigree scoring

**Inputs.** A candidate is available in a component when it has an observed site with likelihood spread above `INFORMATIVE_GL_SPREAD` = $10^{-12}$. An unavailable parent reduces M1 to M0, and M2 to the other parent's raw M1. Availability, information-exponent counts and excursion support use separate informative tests (PS01). At most `markers_per_bin` = 10 markers are kept per painting bin: all of them, or $\text{first} + \lfloor s\,n/10 \rfloor$ for $s = 0, \ldots, 9$. A component is admitted when some selected likelihood triple is non-uniform, and anchoring precedes subsampling (PS02). An information group closes once its running count of selected markers reaches `markers_per_information_block` = 100, with each empty bin counted as one. Then $\eta = \min(3, n)/n$, from `effective_markers_per_information_block` = 3, and $\eta = 0$ without informative markers (PS03). The selector probability $r_b$ uses model distances, fallback tails included, clipped to $[10^{-15}, 0.5]$ (`SELECTOR_SWITCH_FLOOR`). Measured lengths are clipped to the map's knots unless `rate_is_measured`, which defaults to false, and are zero without a map (PS04).

**Projection.** Within the source chain only the initial law is symmetrised. Evidence-free samples take a uniform initial law and unit right weights, and a loss of supported mass is an invariant error (PS05). The projection then symmetrises each bin's marginal. BACKGROUND keeps its unconditioned ALT probability when the conditioning denominator is not positive (PS06). Each bridge is $\text{diag}(\text{stay})$ plus a rank-one off-diagonal term and at most one pivot arm. Its maximum-entropy root is found by at most `bridge_max_iterations` = 128 bisection halvings. A post-check (`bridge_postcheck_tolerance` = $2 \times 10^{-13}$) verifies it; failure stops the stage, with no fallback (PS07). The excursion weight is $k = r_b\,(1 - \min\{1, n/\texttt{source_gap_support_sites}\})$. Here `source_gap_support_sites` = 1 and $n$ is the smaller adjacent-bin informative-marker count (PS08). `null_prior_samples` = 1. A rounding-sized negative null persistence saturates to the uniform kernel; a genuinely negative one raises an error (PS09).

**Child forward.** `mismatch_probability` = 0.01. A bin whose site-ordered product underflows is recomputed in log space. Tempering is $\mathrm{pow}(E/\max E, \eta)$; the power is skipped at $\eta = 1$, and $\eta = 0$ still normalises. The composite weight $\frac{1}{2}$ is fixed (PS10).

**Certified M1 and the panel.** Block-certificate bounds keep called alleles exact and relax uncertain ones to endpoint likelihoods. The bound takes the smaller of two block partitions, with outward padding (PS11). Exact queries cover, in order, the tie class of maximal bounds, the best frontier, the floor frontier and every cell able to reach a local maximum. With chromosome temperatures $t_z = \max(\lceil M_z/100 \rceil, 1)^{1/2}$ (`information_tempering_power` = 0.5), let $t(M)$ be the same function of the genome total of selected markers, $M$. A candidate is retained when its genome M1 reaches $\chi_c = \max \mathrm{M1} + \mathtt{LOG_FLOOR} \cdot t(M)$, with `LOG_FLOOR` = −60. All others receive

$$\tilde{\ell}_{cz} = \hat{\ell}_{cz} - \frac{t_z}{\sum_{z'} t_{z'}} \left( \sum_{z'} \hat{\ell}_{cz'} - \chi_c \right),$$

where $\hat{\ell}_{cz}$ is the certified chromosome-local maximum. The floor is applied exactly once, because it is not idempotent (PS12). Robust screen utilities are diagnostic only (PS13).

The pair bound gives every state with $\pi \geq$ `pair_bound_high_mass` = 0.01 its own group, gives BACKGROUND its own group, and puts all remaining named states in one low group. Emission envelopes use the corners of the transmitted-ALT box, since the site likelihood is bilinear. Float32 bounds are rounded upwards (PS14). A pair enters the panel if, in the full fit or any LOCO fit $w$, $w \cdot \mathrm{bound} \geq \max(w \cdot \mathrm{M0}, \max_r w \cdot \mathrm{M1}_r) - \mathtt{pair_relevance_margin}$, with margin 150, or if the gap is NaN. There is no cap on parents or pairs (PS15). In a component where only one parent of a pair is available, the pair scores as that parent's M1 (PS16).

**Exposure and junctions.** Exposed map length integrates the smaller of the two adjacent bins' exposures over capped measured intervals (PS17). Junction counts come from an exact minimum-switch dynamic programme over unordered pairs of deduplicated trajectories, with costs 0, 1 and 2. Unknown labels are compatible with every trajectory, and callable bins count non-negative labels on both tracks (PS18). The schedule traces the first minimum path, or the last under reversal. Detection is $\min(1, \mathrm{callable}/2B)$. Counts above the child's total are dropped, and zero retained mass gives $-\infty$. Equally parsimonious alternative placements are not integrated (PS19).

**Exclusion e-value.** At a jointly observed site with normalised likelihood triples $x$ and $y$, define $C = \max\{x_0 \max(y_0, y_1), x_1 \max_g y_g, x_2 \max(y_1, y_2)\}$ and $U = \max_g x_g \max_g y_g$. The site ratio is

$$R = \frac{(x_0 y_2 + x_2 y_0)/2}{\max\{C, (1-\delta)C + \delta U\}}, \qquad \delta = \mathtt{genotype_replacement_probability} = 0.01.$$

For chromosome $z$, $E_z$ averages $\prod_{j \in w}(1 - v + vR_j)$ over `bet_sizes` $v \in \{2^{-10}, 2^{-8}, 2^{-6}, 2^{-4}, 2^{-2}, 1\}$ and over windows $w$. The windows are the whole contig and half-overlapping windows of `window_widths_bp` = 1, 5 and 20 Mb. $E$ is the mean of $E_z$ over all configured contigs (PS20). A completed-window lower bound may certify a dyad early, and a threshold-straddling interval raises an error (PS21).

## Pedigree decision and release

**Evidence scale.** Let $\mathcal{T}_{cfa}$ and $\mathcal{H}_{cfa}$ be the training and held-out log-likelihood sums of configuration $a$ for fold $f$. Let $\varpi_{ca} = \log \Pi_{\text{state}(a)} - \log n_{\text{state}(a)}$, where $n$ counts eligible configurations. Let $m_{cfa} \in \{0, -\infty\}$ exclude reciprocal-dominant rows: those naming $p$ as $c$'s parent with $\text{M1}(p \mid c) - \text{M0}(p) \geq \text{M1}(c \mid p) - \text{M0}(c)$, which genetics cannot order. The objective is

$$F(\lambda) = \sum_{c} \sum_{f=0}^{2} [\log \sum_{a} e^{\lambda \mathcal{T}_{cfa} + \varpi_{ca} + m_{cfa} + \mathcal{H}_{cfa}} - \log \sum_{a} e^{\lambda \mathcal{T}_{cfa} + \varpi_{ca} + m_{cfa}}].$$

Folds are chromosome index modulo 4, with at least two chromosomes per fold. The search evaluates a 33-point grid of $\log \lambda$ on $[\log 0.001, \log 100]$ and refines every eligible interior maximum. All optimisations must succeed, and ties go to the value closest to $\lambda = 1$. The scale is applied only when the gain exceeds $10^{-8}$, the optimum lies at least 0.01 from a bound and $\lambda < 1$. Fold 3's test gain is reported only. Setting `evidence_calibration` = "off" gives $\lambda = 1$ (PD03).

**Likelihoods and gate.** Configuration rows are ordered as M0, then M1 by ascending parent, then canonical panel M2 (PD01–PD02). The M0 mixture uses `m0_contamination_probability` = 0.02 and is applied once (PD04). The gate thresholds are `minimum_edge_exposed_bins` = `minimum_pair_exposed_bins` = 1, `minimum_exposed_fraction` = 0.1 and `minimum_exposed_units` = 3, with `unit_morgans` = 1.0. Units are $\max(1, \text{length}/\texttt{unit_morgans})$, or 1 for a non-finite length. A bootstrap-repeated chromosome counts once. An untestable row keeps the M0 state score and gets an identity score of $-\infty$ (PD05).

**Direction and family terms.** For $\Delta \neq 0$, with $\sigma$ the logistic function, chromosome log odds are $\log[(1-e)\sigma(\Delta) + e/2] - \log[(1-e)\sigma(-\Delta) + e/2]$, with `chromosome_error_probability` $e$ = 0.05 (PD06). The configuration term is $\log 2 + \log[(1-\epsilon_p)\sigma(O) + \epsilon_p/2]$, with `pair_failure_probability` $\epsilon_p$ = 0.02. A junction-density order probability, a retained heuristic, only orders graph visits, though this can decide which child of a cycle is displaced (`order_minimum_weight` = 2, `order_variance_offset` = 1; PD07). There are `message_passes` = 4 preparatory cavity passes. Rows naming $p$ as $c$'s parent receive $-\log(1 + e^{O_{pc}^{\text{cav}}})$, where $O_{pc}^{\text{cav}}$ is the cavity log odds that $c$ is $p$'s parent (PD08). The path factor uses `ancestry_path_budget` = 16, moves a boundary tie class that would overfill the panel to the tail, and is then frozen (PD09).

The final solve runs undamped to `final_tolerance` = $10^{-8}$ within `final_max_iterations` = 8192, with progress checks every 512 iterations. A stalled solve is retried damped (`retry_iterations` = 4096, `retry_damping` = 0.5) and then resumed. Non-convergence is fatal for the full fit; a resample abstains but stays in the denominator. Release direction log odds are cavity log-odds differences (PD10).

**Winners, graph and priors.** Two scores tie when their difference is at most $\max(512\,\varepsilon_{\text{mach}} \max(1, |\text{score}|), 2^{-15})$, under `contrast_ulp_factor` and `noise_floor`. Row index never breaks a tie (PD11). The decision score is the marginal state score plus the identity score minus the state's best identity score, and unresolved-state and unresolved-identity reasons stay distinct (PD12). The acyclic step peels cyclic components; tied weakest

children become graph-tie conflicts. Greedy selection then visits children by order probability, confidence and index, and keeps the local row when it is feasible. Otherwise it takes the unique feasible winner from the local state downwards, over `graph_local_search_passes` = 3 passes. This is not global graph optimisation (PD13). The `prior_sensitivity` reselection reuses the family terms and is not repeated in resamples (PD14).

**Units and resampling.** A chromosome is informative when the child's scaled per-chromosome state evidence, without family terms, has a unique winner. That winner is not required to match the selected state. The chromosome contributes the largest exposed units of the child's eligible edges, or 1 for a child without edges (PD15). Bootstrap uses `bootstrap_replicates` = 1000 draws from a PCG64 stream with `bootstrap_seed` = 20260725. Identical vectors are fitted once and counted by multiplicity. Support counts are integers, so scheduling cannot change them (PD16). A precheck needs $N \geq 3$ and 3 informative contigs, and reports attainable support and each tier's minimum contig count (PD17).

**Tiers and views.** The thresholds (configuration bootstrap, parent bootstrap, LOCO) are `tier_b` = (0.7, 0.7, 0.8) and `tier_a` = (0.95, 0.95, 0.95). The direction cutoff is $\log(0.98/0.02)$, from `orientation_reversal_probability` = 0.02. An exact configuration also needs no graph-tie conflict and, for M0, a graph row that stays M0 under every prior. Unresolved statuses rank prior veto, no stable parent, insufficient support; resolved ones rank graph-tie conflict, graph conflict, unresolved direction, unresolved identity (PD18). The complete view keeps its conflict and displacement reasons, and displaced rows are never tier-eligible (PD19). A marginal edge needs the same parent in the local and graph rows, both parent supports, no graph conflict and enough units, but the state itself need not pass (PD20). Candidate sets cover `support_set_coverage` = 0.95 of the within-state softmax of decision scores, completing tie classes at a quantum of $2^{-15}$ (PD21).

**Counts, guard and slots.** Count-release targets are Tier-B children with an unresolved state or fewer named parents than called. The threshold is $\log(2N/\alpha)$, with `alpha` = 0.01. Release needs every candidate scored, enough units, no graph conflict, at most one survivor, agreement with the supported and complete parents, and no withheld edge, and is labelled as exclusion-based evidence (PD22). The guard uses `minimum_family` = 3, so at least two tested members, and `member_support` = 0.75. It contracts the edge set to a fixed point in at most $E + 1$ passes, runs count release with the withheld set, and verifies both sets. Guarded views clear the affected calls but keep their candidate lists (PD23). Slots order parents by descending observed-site counts, compared contig by contig, with names breaking exact ties (PD24). Relative-proxy contrasts are diagnostics that no release reads (PD26).

**Cost.** Sources cost $O(NBS^2)$, transmitted ALT $O(NMS)$, the pair bound $O(R^2BG^3)$ for $R$ retained candidates and $G \leq S + 1$, and exposure $O((E/2 + T)BS(S + 1)/2)$ for $T$ admitted pairs. The decision has no intrinsic $K$ term and runs at most $1000 + C$ fits.

### Founder allele refit

Evidence uses the founder-construction view, so QC-flagged samples are unobserved. A sample scores at a site only when both of its tracks carry released singleton rows, and sites with no scoring sample are unchanged (FR01). Each site gets `coordinate_sweeps` = 3 sweeps. A sweep has a flip phase, then a two-row phase in which swaps and pair flips are restricted to co-painted rows. A move qualifies when its gain is positive and at least `minimum_gain_nats` = 5 per changed cell. A round applies qualifying moves in decision order, skipping any that share a row, or a row co-painted with one of theirs, with a move already applied. A row changes at most once per phase and sweep, and a sweep without changes ends the site. Qualification and rechecks use the same direct two-row gain (FR02).

Tie classes chain gains within $\max(2^{-16}, (n_i + n_j)2^{-18})$ nats, where $n$ counts scoring carriers (`REFIT_TIE_QUANTUM`, `REFIT_TIE_QUANTUM_PER_CARRIER`). A class wider than `WIDE_QUANTA` = 4 quanta keeps value order; otherwise row or pair index decides. Reversal reorders a class but never changes its membership (FR03). Interacting candidates in one tie class are applied hypothetically in both orders; if either then fails to qualify, all rows involved become unknown in one pass, the round is rescored, and those alleles cannot be filled later (FR04). A fill applies only to alleles unknown in the painted panel. It needs a gain of 5 nats and at least `fill_min_supporters` = 2 carriers whose reads favour the allele by more than $2^{-16}$. Fills are decided against the post-sweep state and applied together (FR05). The refit panel goes to family phase and export; pedigree keeps the pre-refit founders (FR06). The cost per site and sweep is $O(N + K^3 \log K)$. In the worst case, two-row gains cost $O(K^4)$, and tie resolution $O(K^4)$ flips and $O(K^5)$ swaps per phase. This is exception 0e (FR04), carried for review in release 4.1.

## S5. Family phase, recombination maps and released outputs

### Family phase

Notation follows the main text. Parameter names are those of the generated parameter reference; values are 4.0 defaults.

**Emissions, released sites and state alleles (FP01, FP13).** The pair emissions $e_{ib}(x, y)$ are the painting's binned emissions, read from its cache or recomputed identically with its `robustness_epsilon` (0.01), `emission_log_floor` ($-50$ nats) and design effect $1 + (n_{ib} - 1)\rho_s$. Here $n_{ib}$ counts informative sites of sample $i$ in bin $b$ and $\rho_s$ is `emission_site_correlation` (0.1). Both solves share these pre-refit emissions. Released sites are refit panel positions inside a painting bin; other sites keep missing calls. A named state's allele $\hat{h}_{xj}$ exists only where every row of its class has a called refit allele and all agree; BACKGROUND has none. Components without painting bins are not solved. They keep their geometry (fallback bins of `EMPTY_COMPONENT_BIN_SITES` = 100 panel sites), but their calls and chain sources stay missing.

**Edge rates (FP02).** $\theta_b^{\ell} = 1 - \sqrt{\psi_b}$, where $\psi_b$ is the painting's pair stay probability, clipped to $[10^{-15}, (S-1)/S]$. The other clips are $[10^{-12}, 1/2]$ for $\theta_b^{m}$, $[10^{-12}, 1/2]$ for $\alpha_b$ and $[10^{-6}, 1-10^{-6}]$ for $\beta_b$. `not_copy_rate_per_bp` $\rho_{\perp} = 10^{-8}$ and `not_copy_mean_bp` $\mu_{\perp} = 5 \times 10^{4}$ give $\omega \approx 5 \times 10^{-4}$. One Haldane implementation (via `expm1`) serves every map distance. `[family_phase] protected_intervals` is a TSV (`chrom, start, end, reason`; 1-based inclusive). Ordinary rows protect their span; a `reversed_segment` row protects its two endpoints. An edge whose closed centre-to-centre interval overlaps a protected interval gets $\theta_b^{m} = 1/2$.

**Local gains (FP03, FP25).** Because $\perp$ entry draws a uniform label, $J$ is a sum of per-sample terms, each reading only that sample's labels and sources. A move's gain is the accurately rounded sum of after-minus-before terms over exactly the samples it changed (the moved sample and its descendants); no change gives exactly zero. This local gain decides acceptance. The full $J$, recorded per sweep, is a verification quantity that may round differently near a threshold. A test-only reference path compares the two and never alters production arithmetic.

**Block recursion, ties and order (FP04, FP05).** The block transition is contracted in three stages (previous source of homologue 0, previous source of homologue 1, previous gauge). This costs $O(B(n_0^2 n_1 G + n_0 n_1^2 G + n_0 n_1 G^2))$, plus $O(B\,|\mathrm{desc}|\,S^2)$ for descendant emissions. Only gauge 0 is allowed at the first bin. In each stage a candidate replaces the incumbent if its score beats the retained best by more than `tie_quantum` ($10^{-9}$ nats), or lies within it and is the slot's current state (for the gauge, the unchanged gauge). The retained argument's score, not the numerical maximum, continues the recursion. At the final bin, the current sources with gauge 0 win ties. Samples are ordered by depth (longest ancestor chain), then index; the graph must be acyclic. These orders fix the Gauss–Seidel updates.

**Initialisation and sweeps (FP06, FP07).** The first pass solves every sample in index order with both slots free and no child terms, from all-zero labels. The second pass solves samples with a released parent in ancestor order, initially copying parental homologue 0. Neither pass applies a gain test. Sweeps keep a block only if its local gain exceeds `gain_quantum` ($10^{-6}$ nats), restoring it otherwise, for at most `max_sweeps` (30) sweeps. A component is settled when the final sweep of both solves accepted nothing; the recombination summary reports this as `family_phase_converged`.

**Pairing moves (FP08).** Candidates are samples with no released parent and at least one child, at every bin edge where both of their homologues change label. They are enumerated once per pass in pedigree then bin order. A move that changes nothing is skipped. A move is kept if its gain exceeds $\max(\log[\tau_s/(1-\tau_s)], \texttt{gain_quantum})$, which is 6.907 nats at $\tau_s = 0.999$. Passes with kept moves trigger more sweeps, for up to `max_sweeps` passes. At that cap, a final non-keeping pass measures ambiguity in the final state. A pairing is ambiguous when the alternative's gain exceeds $-\log[\tau_s/(1-\tau_s)]$.

**Convention solves (FP09).** The mirrored solve reverses three orders: visits within a depth, the scan direction for non-incumbent ties, and descendant order during ascent. It keeps the incumbents, gauge preferences, thresholds and first-pass order. Neither solve initialises the other. The default solve's orientation is released even when the run's reporting convention is reversed.

**Identity junctions (FP10).** A candidate is an edge $j$ and labels $\{a,b\}$ at which at least `junction_min_samples` (2) distinct samples have a free track, or a track not-a-copy at $j-1$ or $j$, switching between $a$ and $b$. A copied homologue's switch belongs to its parent or is a crossover, so it is not counted. A sample carrying $\{a,b\}$ on both sides of $j$ is pinned only under three conditions. A linked homologue must copy parental homologue $v$ at both $j-1$ and $j$. The parent must not itself carry both labels. On each flank of up to `junction_pin_bins` (10) bins, stopping where the parent's labels change,

$$\sum_{b'} [e_{pb'}(x,y) - \max_{y'} e_{pb'}(z,y')] \geq \texttt{junction_pin_nats}\ (10),$$

with $x$ the label on homologue $v$ and $z$ the exchanged label. The partner $y'$ is free because two rows exchanged together can fit the same unphased reads. Children never pin: a child's copy excludes the alternative only through a meiosis switch exactly at $j$, whose odds depend on the interval and on uncertainty the children share. Withheld joins of both solves are combined and cannot be bridged.

**Painting ties and forced breaks (FP11, FP12).** At each boundary between a sample's released painting chunks, direct and swapped continuations are compared lexicographically by (label changes, allele-visible changes). The boundary is a tie unless direct continuation is strictly better. It is mapped to the next site's bin and kept only at internal bins of samples with no released parent. Forced breaks combine tie bins, withheld joins and ambiguous-pairing edges (applied to the parent and all its children) from both solves. Only tie-only cuts can be bridged.

**Panel-allele uncertainty (FP14).** The prior is the logit of the chromosome-wide share of allele 1 among called refit cells, clipped to [0.01, 0.99] (`PRIOR_BOUNDS`). Carriers of $x$ at site $j$ are samples whose optimal label pair contains $x$; samples flagged by sample quality control are excluded, though they are still phased. Each carrier adds $\ell(g_1) - \ell(g_0)$, where $\ell(g) = \log[(1-\varepsilon)P(r_{ij}, a_{ij} \mid g) + \varepsilon/3]$, $\varepsilon$ is the painting's `robustness_epsilon`, and $g_u$ is the carrier's dosage if $x$ carries allele $u$ given its partner's selected allele; an $(x,x)$ carrier compares dosage 2 with 0. With $\gamma_{xj} = \text{logistic}(\text{prior} + \Lambda_{xj} - \Lambda_{xj}^{(i)})$, where $\Lambda_{xj}$ sums the carrier terms, $d_{xj}$ is the disagreement of $\gamma_{xj}$ with the selected allele, capped at 1/2 (`FLIP_CEILING`) and zeroed below $10^{-15}$. The sample's own direct term $\Lambda_{xj}^{(i)}$ is removed for the labels of its optimal pair. Documented approximations remain. This is not a cross-fitted refit, and the sample can still influence carrier and panel selection. The two copies flip independently even for $(x,x)$, and correlated carrier identity errors remain a failure mode.

**Posterior, site model and calls (FP15–FP17).** Forward–backward uses the Viterbi factors, starts the gauge at 0 and enforces any anchor in both directions. States with log posterior $\leq -700$ are dropped without renormalisation. Descendant sums follow a fixed (depth, index) order in both solves. Chain probabilities stay in float64 until the release test. Settings are refused unless $\tau_s < (1-5\delta/6)/(1-2\delta/3)$, the orientation certainty of a fully determined heterozygote (0.99983 at $\delta = 10^{-3}$). They also require $\tau_p \leq \tau_s$, all thresholds in $(1/2, 1)$ and $\delta \in (0, 1/2)$. `site_deviation` ($\delta$) and `set_posterior` ($\tau_s$) are working values chosen on validation simulations, not estimated error rates. Genotypes use the argmax dosage,

with the lower dosage winning exact ties, at the inclusive `genotype_posterior` ($\tau_g$ = 0.9). Heterozygotes are phased at the inclusive `phase_posterior` ($\tau_p$ = 0.99). The order is (1,0) only if $P_{10} > P_{01}$, otherwise (0,1). Calls carry both alleles or neither.

**Gauge-sample sets (FP18).** These rules apply to samples with neither a released parent nor children. The anchor is the first bin of a partition whose optimal labels differ and whose label pair is heterozygous with posterior ≥ $\tau_p$. Certainty is computed over whole bins up to the next forced break. The set stops at the first later bin with differing optimal labels, label heterozygosity ≥ $\tau_g$ and certainty < $\tau_s$. Undefined certainty (no confidently heterozygous site) does not stop it. Pre-anchor bins join the anchored partition; each stop opens a new partition.

**Children-anchored gate (FP19).** For a sample without released parents but with children, a tie-only cut is bridged when the certainties at the nearest informative bins before and at-or-after the cut both reach $\tau_s$. A cut lacking such a bin on one side is kept. A break kept by either solve is kept.

**Agreement (FP20).** Breaks are combined, and a set also starts wherever the relative orientation of the two solves changes between consecutive phased heterozygotes. The first site is never a break.

**Chain release (FP21).** Resets fall at the component start, wherever $\theta_b^m \geq 1/2 - 10^{-12}$, and at the bins of either individual's agreed set starts. A start inside a bin (after its first released site) blocks that bin and resets the next. $\kappa_{eb}(v)$ excludes not-a-copy mass. For a parent without a released parent, alignment uses the bins of each reset segment where the default label pair is heterozygous and the unordered pairs match. If all those bins are swapped, the mirror is flipped; if only some are, the segment is blocked. Bins whose aligned parental labels still differ are blocked. Supports are stored as float32, zero where unreleased.

**Storage and product (FP22–FP24).** Resident or windowed storage, fixed 4,096-site chunks and worker count do not change calls, phase sets or chains. Each product binds the exact parent-edge digest it was conditioned on. Painted-span coverage is diagnostic, and recombination does not read it.

**Cost in K.** A sweep costs $O(B(NS^3 + S^2 \sum_i A_i))$, with $A_i$ one plus $i$'s descendant count. A pairing move costs $O(n_c BS^3)$ for $n_c$ children. Release costs $O(NB(S+2)^3 G)$ for posteriors, $O(LS^2)$ per sample for the site model, and $O(EB)$ for chains. Emissions take $8NS^2 B$ bytes. Every term is at most cubic in $K$ ($S \leq K + 1$).

## Recombination maps

**Breakpoints (recombination RC01, RC02).** `[recombination] order_breakpoints` uses the protected-interval TSV format; the two settings are independent. Ordinary intervals (in file order), then reversed-segment endpoints as point intervals, are made half-open as $[\text{start}, \text{end} + 1)$. Each is widened cumulatively to every overlapping painting bin and removed from the map axis, with a new segment after it. Inversion interiors are not masked automatically, and calls inside reversed segments are flagged. A chain bin spanning two segments or a masked zone is invalid. Segment changes reset chains.

**Following (recombination RC03).** A marker is a released bin in a valid segment; stretches restart at any chain or segment reset. A called interval is $(c_{m-1}, c_m]$, between bin centres. Its reported interval is the inclusive bp range of both bins, so a change inside either bin is not mislocated. The joint lower bound $\max(0, \bar{\kappa}_l + \bar{\kappa}_r - 1)$ holds under any dependence. For each meiosis the stage counts released, followed and unresolved bins.

**Conditional map (recombination RC04, RC05).** Map bins start every `bin_bp` (1,000,000 bp) from the first site, with extra edges at the two sites flanking each segment change. $\lambda$ is the difference of cumulative Morgans interpolated at bin centres. Map knots between the end sites are retained, so hotspots between SNPs keep their mass. The fallback rate `[genetic_map] rate_cm_per_mb` (default 5 cM/Mb) applies without a map and beyond its knots. An odd interval of zero genetic length contributes one crossover: the data override the map's zero prior. Edge sums follow a fixed order. The columns are:

- `rate_cM_per_Mb` is $10^8 X_t / W_t$, missing where $W_t = 0$.
- `covered_bp` is the union of all meioses' informative intervals within the bin.
- `cM_on_covered_bp` is rate × covered bp / $10^6$; the cumulative sum counts missing values as zero.
- `effective_meioses` is $W_t$ divided by the map-bin width.
- The called rate divides called crossovers (spread by genetic mass, else uniformly in bp) by callable exposure.

`rate_is_measured` does not alter the calculation.

**Junction diagnostic (recombination RC06).** Overlapping or touching reported intervals form loci, and a locus is admitted when at least `minimum_parents` (3) distinct parents have calls in it. Each edge is scored by the parity of its calls in the locus. It is decidable if it has a call there or a callable span covering the calls' extent. A parent with $\eta_1$ odd of $\eta$ decidable meioses contributes $\log\left[\frac{1}{2} q^{\eta_1} (1-q)^{\eta - \eta_1} + \frac{1}{2} q^{\eta - \eta_1} (1-q)^{\eta_1}\right]$. Here $q$ is the probability of a real transmission change, and the mixture is over a frame flip (prior 1/2) shared by the parent's children. Three statistics follow.

- **Mixing.** The maximum over 500 values $q \in [0.001, 0.5]$, minus the maximum over $q \le q^{\max}$. The ceiling is $q^{\max} = q_{\text{pt}} + r_{\text{bg}} - 2 q_{\text{pt}} r_{\text{bg}}$, with `hotspot_switch_probability` $q_{\text{pt}} = 0.05$ and $r_{\text{bg}}$ the Haldane fraction across the locus.
- **Pairing.** For each locus pair, a discordance mixture over 50 values $\phi \in [0.005, 0.25]$, among parents decidable at both loci, minus the larger independent switching maximum. The best partner is kept, minus $\log(\text{locus pairs tested})$.
- **Flag.** A locus is flagged when the larger statistic reaches $\log 100$ (`flag_log_likelihood_ratio`).

The documented rationale is that no locus gives real crossovers in more than 5% of meioses beyond background, with flags at 100:1. Ties take the first maximum, or the last under reversed conventions. The ratios are conditional on the calls, not calibrated odds, and never mask the map.

**Stage and cost (recombination RC07).** The stage refuses chains conditioned on another parent-edge digest. Its cost has no intrinsic $K$ dependence: $O(EB)$ following over the $B$ chain bins, $O(EB_m + \text{calls})$ aggregation over $B_m$ map bins, and a detector cost of $O(\text{loci}^2 E)$ plus $O(\text{loci}^2 \times \text{parents} \times \text{grid})$.

### Released outputs

**Site rows (OU01).** By default (`[export] full_length = true`) export writes every input biallelic SNP reaching the site filter, in position order. Multiallelic, non-SNP and duplicate-position records dropped by the loader are absent. Kept sites have FILTER `PASS`; removed sites list their rules in r1–r5 order; a kept-only export writes `.`.

**Founder VCF (OU02–OU04, OU06).** Column `cC_Hk` holds row $k$ of release segment $C$ as a haploid GT, and is `.` outside the segment's panel span. Upstream (OU03), a called allele is withheld unless an eligible painted carrier supplies observed read support, using refit calls. At sites removed only by the informativeness rule r5, unwithheld rows receive the strict majority of summed REF and ALT allele depths (GT only where AD is unusable). Ties stay `.`, and the fill carries no founder-specific evidence. Sites removed by r1–r4 stay `.`. Founder-identity flags mark duplicate-partner changes between blocks, and are diagnostic only.

**Sample VCF (OU05, OU09).** Columns are in input order. A set starts at a component's first kept site and at the site after each released break; its PS is that bp position, which need not be heterozygous. Inside a set, homozygous and missing cells also use `|` and PS. A frame-undetermined heterozygote is written `/` with ascending alleles (missing first) and PS `.`, without ending its set. Sites outside components are unphased, removed sites are `./.:.`, and there is no r5 fill.

**Recombination tables (OU18).** For each contig there are crossover, meiosis, map, coverage, called-coverage and junction tables, plus a summary, with call intervals $(\text{left}, \text{right}]$. The plot and tables are views of one conditional map.

**Cost.** The sample VCF costs $O(NL)$ and the founder VCF $O(L \times \text{columns})$ over $L$ written sites, at most quadratic in $K$.

## S6. Simulation, evaluation and computation

### Simulation settings and constants

Table S1 lists every simulation setting with its default. Settings in `[simulation]` and `[simulation.reads]` describe the generating model only; inference keeps its own read model and map. The order of `[input] contigs` is part of the random streams, and every contig's template must hold the same even number $H$ of haplotypes.

| Setting | Default | Allowed values | Meaning |
|---|---|---|---|
| `seed` | 400 | integer or `"random"` | Seed; `"random"` takes eight OS-random bytes once at initialisation and |

| Setting | Default | Allowed values | Meaning |
|---|---|---|---|
| | | | records the realised value |
| `templates` | none (required) | directory of `<contig>.npz` | `positions` int64[$L$], `allele_probabilities` float64[$H$, $L$, 2] |
| `cohorts` | [20, 100, 200] | each $\geq 2$ | Cohort sizes; cohort 1 descends from the founders |
| `depth` | 5.0 | finite, $> 0$ | Mean depth $\lambda$ |
| `generating_rate_cm_per_mb` | 5.0 | — | Fallback rate $\gamma$ (5 cM/Mb $= 5 \times 10^{-8}$ Morgans per bp) |
| `generating_map` | none | cumulative cM map | Generating map, independent of inference |
| `backcross_fraction` | 0.0 | $[0, 1]$ | Per-child backcross probability in eligible cohorts |
| `backcross_start_cohort` | 3 | $\geq 2$ | First cohort that may backcross |
| `observe_cohorts` | [] (all) | unique, generated cohort numbers | Cohorts eligible for sequencing |
| `observed_fraction` | 1.0 | $(0, 1]$ | Sequenced fraction $\theta$ of eligible individuals |
| `pedigree` | none | TSV: sample, parent1, parent2, sex, generation, observed | Explicit pedigree; replaces `cohorts` and random matings; `backcross_fraction`, `observe_cohorts` and `observed_fraction` must keep their defaults |
| `reads.error_rate` | 0.02 | $(0, 0.5)$ | $\varepsilon$ |
| `reads.depth_cv` | 0.0 | finite, $\geq 0$ | $v$ |
| `reads.heterozygote_alt_probability` | 0.5 | $(0, 1)$ | $b$ |
| `reads.dropout_fraction` | 0.0 | $[0, 1)$ | $\phi$ |
| `reads.read_length_bp` | 0 | integer $\geq 0$ | $\ell$; $\ell > 0$ selects fragments |
| `reads.fragment_mean_bp` | 375.0 | $> 0$ when $\ell > 0$ | $\mu_{\text{frag}}$ |
| `reads.fragment_sd_bp` | 150.0 | $\geq 0$ when $\ell > 0$ | $\sigma_{\text{frag}}$ |

Table S2 lists the internal constants, which are not user settings.

| Constant | Value | Role |
|---|---|---|
| `MUTATION_RATE_PER_BP` | $10^{-10}$ | $\mu$, per bp per meiosis |
| `MINIMUM_FRAGMENT_BP` | 50 | Lower clip of insert length |

| Constant | Value | Role |
|---|---|---|
| FRAGMENT_SD_LIMIT | 6 | Upper clip $z_{\max} = \lceil \mu_{\text{frag}} + 6\sigma_{\text{frag}} \rceil + 1$ bp |
| FRAGMENT_QUANTILES | 100,000 | Quadrature points for the expected bases covered per fragment |
| WINDOW_BP, WINDOW_SHIFT_BP | 100,000; 50,000 | Template windows and stride |
| TEMPLATE_HAPLOTYPES | 6 | Linker path budget |
| UNDECLARED_LENGTH_BP | 300,000,000 | Contig length when the VCF header gives none |
| SIMILARITY_CUTOFF | 50 (%) | Junction mismatch cutoff |
| NODE_PENALTY, EDGE_PENALTY | 10; 10 | Penalties applied after each extracted path |
| GAP_SCORE, EDGE_DIVISOR | 1; 3 | Score of non-overlapping windows; edge weight = score / 3 |
| BISECTION_STEPS | 60 | Inversion of merged information |
| PATH_TOLERANCE, CANCELLATION_TOLERANCE | $10^{-9}$ | Path reconstruction; exact information cancellation |

The `haplotypes example` generator (OU22) writes three contigs of 1,200 evenly spaced sites over 12,000,000 bp with six independent Bernoulli haplotypes, cohorts (20, 30, 30) and depth 5.0; it demonstrates software wiring only. Two simulation configurations are shipped: `configs/simulation.toml` (22 contigs, chr1–chr20, chr22 and chr23; seed 400; cohorts [20, 100, 200]; depth 5.0; generating and inference fallback rates 5.0 cM/Mb) and `configs/simulation_backcross.toml` (the same contigs; seed 6202; cohorts [20, 30, 30]; `backcross_fraction` 0.5 from cohort 3). TO FILL: list the configurations, seeds and read-model settings of the simulations reported in the Results.

## Generating-model details

**Templates (OU20, OU21).** Windows start every 50,000 bp from 0 to the contig length and cover $[\text{start}, \text{start} + 100,000)$. For rows $k$ and $k'$ of consecutive windows, the mismatch percentage $\Delta$ is 100 times their expected allele mismatches, $\sum_j P_k(0)P_{k'}(1) + P_k(1)P_{k'}(0)$, divided by the number of shared discovery sites; the similarity $\sigma = 0$ if $\Delta > 50$, otherwise $2(50 - \Delta)$, and the junction score is $100(\sigma/100)^2$. Paths are found by reverse dynamic programming, the first candidate winning exact ties. After each path, every row's node score in a used window falls by 10 times its similarity to the used row, with similarity $1 - \min(1, 2\delta/n)$ for $\delta$ expected mismatches over the window's $n$ discovery sites, normalised by the square roots of the diagonal, and each internal used edge falls by 10. Overlaps are merged with signed information $I(p) = \text{sign}(p - \frac{1}{2})(1 - H_2(p))$, where $H_2$ is the binary entropy, combined as $(I_1 + I_2)/(1 + I_1 I_2)$ and inverted by bisection. A contig with fewer than two informative windows gets no template. The linker is compiled without fast-math reassociation (ledger L14), which matters only when templates are rebuilt.

**Founders (OU23).** Each haplotype makes one vector draw for its exact ties, in site order, and only when ties exist.

**Pedigrees (OU24, OU25).** In the cohort design, the master stream's first draw seeds the structural generator. Each cohort's structure is built before its seed block is drawn. In a backcross-eligible cohort every child consumes one structural uniform draw: below `backcross_fraction`, the first parent is drawn uniformly from the previous cohort and the second uniformly from that parent's two parents; otherwise two distinct previous-cohort members form an ordered pair (first parent gives the maternal haplotype). In an explicit table, founders have depth 0 and others one more than their deeper parent; individuals are simulated by depth in table order, without a structural draw, and the founder count must equal $H/2$. Generated identifiers (`Founder_i`, `F<cohort>_<i>`) are truth labels only.

**Meiosis (OU26).** Without a map, gaps of $\lceil \mathrm{Exp}(\text{mean } 1/\gamma) \rceil$ bp run from the first marker until past the last. With a map, the Poisson mean is the distance from first to last marker, including fallback-rate tails beyond the map's knots. A zero distance or rate consumes no crossover draws, and a marker exactly at a crossover takes the new homologue. Mutation gaps are $\lfloor \mathrm{Exp}(\text{mean } 1/\mu) \rfloor$ bp; each event flips the first marker at or after it, and duplicate indices flip once. The child stream draws the maternal gamete, then the paternal.

**Reads (OU27, OU28).** The independent-site sampler draws, in order: depth factors (only if $v > 0$), the full $N \times L$ Poisson array, dropout starts (only if $w > 0$), then binomial ALT counts per dosage class 0, 1, 2, skipping empty classes. The fragment sampler draws factors once. Then, per sample, it draws a Poisson fragment count over the span from $z_{\max}$ bp before the first marker to the last marker, uniform starts, lengths and homologues, followed by per-read imbalance and error draws. Dropout starts are drawn after all samples. The start rate is $\lambda u_i/(1 - w/L)/\bar{c}$, where $\bar{c}$ is the mean of $\min(z, 2\ell)$ over insert lengths $z$ at the quadrature points. Both samplers consume whole-contig arrays, so splitting by sample would change the draws.

**Streams (OU30).**

| Stream | Construction | Draws |
|---|---|---|
| founders | PCG64, `SeedSequence([seed, 2000000])` child $c$ | Tie resolution on contig $c$ |
| master | PCG64, seed | Structure seed (cohort design), then per cohort or depth a contig-major $C \times n$ child-seed block |
| structure | Mersenne Twister seeded by `master.integers(0, 2^31)` | Parent pairs and backcross choices |
| child | PCG64, child seed | Maternal then paternal gamete |
| observation | PCG64, `SeedSequence([seed, 3000000])` | Sequenced subset |
| reads | PCG64, seed + 1,000,000 | One seed per contig, for every contig |
| contig reads | PCG64, contig read seed | Whole-contig reads |

Here $C$ is the number of contigs and $n$ the cohort size.

**Truth (OU29).** Per contig: positions; founder haplotypes int8[$H$, $L$] (consecutive row pairs form the diploid founders); sequenced samples' alleles int8[$N$, $L$, 2] (maternal, paternal); ancestry chunks with integer $[\text{start}, \text{end})$ bounds and maternal and paternal founder haplotype ($-1$ for none); and one row per gamete of a sequenced child (child, parent, slot, cohort index, float crossover positions). The run manifest holds the state, requested and realised seeds, contigs, source identity, sequenced sample identifiers, and the sequenced and generated pedigree tables. `simulation/truth_pedigree.csv` summarises the sequenced samples' true parents.

## Metric definitions and edge cases

**Selection (OU31).** The default `available` mode scores the latest founder product of each contig plus any completed pedigree, family phase and recombination products, listing missing downstream products as skipped. `all` scores every founder product present; explicit stages fail if a product is missing. A complete simulation manifest is required. The pedigree comparison stays genome-wide when contigs are restricted.

**Founder panels (OU32, OU33).** A stage's components must not overlap and must lie on truth markers, and every truth marker counts. If $K_c > H$, $K_c - H$ rows stay unmatched and their calls are reported as unmatched called alleles; if $K_c < H$, each unmatched founder costs $L_c$ missing alleles. When several assignments are equally optimal, assignment-derived counts and strata can differ between them; a pure row permutation changes no count. Per-million rates are null with a zero denominator. In the segment path, an equal switch or stay cost keeps the current label, and the first terminal minimum wins. An infinite switch cost reproduces the nearest-founder count. Component-wise counts can fall when a component is cut, because labels re-match on each piece; the split-charged cost rises by at most $\kappa$ per row.

**Ancestry and strata (OU34).** Founder $f$ is represented at site $j$ if $c_{fj} \geq 1$, ignoring unknown ancestry. Chunks are half-open, except that the chunk ending at the last marker includes it. Per stratum, missing alleles are truth alleles minus correct and wrong matched calls, so they include uncalled alleles, unmatched founders and uncovered markers.

**Identity continuity (OU35).** Windows are consecutive blocks of 100 truth markers, the last possibly partial, each centred on its middle marker. A row is labelled in a window when it has $n \geq 16$ calls there, the best founder has at most $\max(1, \lfloor 0.05n \rfloor)$ mismatches, and the second best has at least 2 more. An assertion between consecutive labelled windows $w_1 < w_2$ is direct if $w_2 = w_1 + 1$ and gap-spanning otherwise. A window's copy class is $\min(2, \lfloor \text{median } c_{fj} \rfloor)$. A correct assertion takes the poorer class of its two windows; a wrong one takes the departing founder's class in $w_2$. An assertion is flagged when a pipeline identity flag of its row lies in (centre of $w_1$, centre of $w_2$]; flagged assertions are counted separately, never as errors avoided. A founder's eligible windows have median copy count $\geq 1$ and Hamming distance $\geq 3$ to every other founder. Consecutive eligible windows at most 3 apart are covered when one row labels the founder in both and labels no other founder in between. Coverage is also reported without flagged rows. Splitting rows or dropping assertions can raise the correct share without raising coverage.

**Phase (OU36; definition `released_phase_v3`).** A new set starts at each component change and after each released break (boundary $j$ lies between sites $j$ and $j+1$). Sites outside components belong to no set. A scored site has both alleles called, the true allele sum, a true heterozygote, set membership, a called heterozygote, and a displayed phase that is not undetermined. Released sets need only a called heterozygote released phased: a call 0|1 against truth 0/0 gives one genotype error, one released set and no scored set. N50 is 0 without a positive span; the phased share is null with a zero denominator.

| Orientations within one set | Scored sites | Site errors | Switch events | Long switches |
|---|---|---|---|---|
| `0 0 0 1 1 0 0 0 0 1` | 10 | 3 | 3 | 0 |
| 30 zeros, 2 ones, 25 zeros, 21 ones | 78 | 23 | 3 | 1 |
| Alternating at every site, length $n$ | $n$ | $\lfloor n/2 \rfloor$ | $n-1$ | 0 for runs below the long-run threshold |

Parent groups count released post-guard Tier-B edges into each sample, not true or cohort-derived parents.

**Pedigree (OU37).** The expected state is the number of a sample's true parents that were sequenced. An unresolved call (state $-1$) is never state-correct. Extra edges are called minus correct; missing edges are true minus correct.

**Map (OU38).** Edges are keyed by (parent, child) names over the gametes of sequenced children; an inferred edge that is not a true gamete is counted and skipped. A true edge without crossovers still contributes expected counts, exposure, calls and widths. Hit rates and median widths are null without calls. Called and reported intervals both count a hit when a true crossover lies in $(\text{left}, \text{right}]$.

## Cost exemptions and exceptions at $K = 30$

| Decision | Retained behaviour | Cost | Meaning at $K = 30$ |
|---|---|---|---|
| 0 | Rare context-coded local-selection screening | $O(K^5)$ per round | $K^5 = 24,300,000$, an algebraic factor, not a measured count or time |
| 0b | Ranked assembly and refinement seed refits (`ranked_refits` cap) | Fixed admission cap | The cap does not grow with $K$, so candidates can be omitted at larger $K$; costed as 9 ranked refits and at most 2,048 shadow scores per member |
| 0c | Bounded discovery and local-selection searches | Discovery $O(NLK^4)$; selection backstop $O(NLK^3(K+L))$; context snaps $O((K+L)^2)$ passes | $K^4 = 810,000$, $K^3 = 27,000$; full cost depends on $N$, $L$ and the search path |
| 0d | Optional pairwise IBD | $O(N^2 L)$ | Independent of $K$; 3,123,750 unordered |

| Decision | Retained behaviour | Cost | Meaning at $K = 30$ |
|---|---|---|---|
| | | | pairs at $N = 2,500$; off by default |
| 0e (AJ17) | Assembly joins' structural screening | Quartic trend in $K$ of the direct query count on a finite grid | Analytic worst case; emissions ≈18.6 GB at $N = 2,500$, $m = 2,000$ |
| 0e (RS05, RS07) | Refinement window fixing | Bounded only by the work account | Analytic worst case; polish ≈3.7 GB at $N = 2,500$; provisional |
| 0e (PR17) | Painting label-continuity checks | $O(K^4)$ checks | $\approx 10^{11}$ gain operations at $N = 2,500$ with 2,000 links |
| 0e (FR04) | Founder-refit ties | $O(K^4)$ flips and $O(K^5)$ swaps per phase; $O(K^4)$ over $K$ rounds via the direct two-row gain (L10) | $K^4 = 810,000$; $K^5 = 24,300,000$ per site and phase; observed at most 3 rounds and 2 tie members |

None of the 0e worst cases binds on the validation inputs; all four are scheduled for review. The join size $m$ is as written in the cost table, whose symbol list does not define it.

### Tie quanta and execution units

For the shared decision kinds, $q_0 = 2^{\lceil \log_2 (16v) \rceil}$ with noise motions $v$ of $5.5 \times 10^{-5}$ (NLL, improvement), $10^{-5}$ (allele step), $5 \times 10^{-6}$ (sample), $2 \times 10^{-6}$ (cell) and $10^{-6}$ (probability). The relative factor $a$ is $2^{-30} \cdot 2^{10}$ for NLL, $2^{-36} \cdot 2^{10}$ for sample, and 0 otherwise. Assembly decisions use their own floors and factors, and fixed policies use a declared width (EC30–EC32). The main-text execution mechanisms correspond to the dependency graph and barriers (ENG-01, ENG-03), run modes (ENG-04), identities and resume (ENG-06–ENG-08), the append-only container (ENG-14), the CPU budget and pools (ENG-20, ENG-21), contig lanes (ENG-23), the memory governor (ENG-25) and the kernel cache (ENG-27).

## S7. Supplementary tables

**Supplementary Table S1.** Kernel ladders of release 4.0. Each probe times one kernel on direct synthetic inputs, on one thread, after a discarded first call; slopes are fitted log–log exponents with standard errors. A K slope at or below 3.3 is accepted as at most cubic; "exempt" marks a stage under a decided cost exemption, and "size proxy" an operational kernel whose size stands in for K and N. Source: docs/developer/costs.md.

| Stage | Probe | What the probe covers | K slope (K = 5–30, N = 200) | N slope (N = 250–2,500, K = 15) |
|---|---|---|---|---|
| evidence | evidence.calibration_objective | sample objective/gradient at 230 depth-alt cells; no intrinsic K | -0.00 ± 0.00 (≤ cubic) | 1.00 ± 0.00 (measured) |
| evidence | evidence.likelihoods | read-model likelihood application N×256; no intrinsic K | -0.00 ± 0.00 (≤ cubic) | 0.98 ± 0.02 (measured) |
| evidence | evidence.quality_genotypes | genotype expectations, default 60 iterations, N×256; no intrinsic K | -0.00 ± 0.00 (≤ cubic) | 1.12 ± 0.01 (measured) |

| Stage | Probe | What the probe covers | K slope (K = 5–30, N = 200) | N slope (N = 250–2,500, K = 15) |
|---|---|---|---|---|
| relatedness.families | relatedness.screen | all singleton family screen; F=N; no intrinsic K | -0.02 ± 0.01 (≤ cubic) | 2.00 ± 0.00 (measured) |
| relatedness.families | relatedness.terms | AF and family terms; no intrinsic K | -0.01 ± 0.01 (≤ cubic) | 1.02 ± 0.04 (measured) |
| relatedness.pairs | relatedness.pairs | all pairs, 16 thinned loci; no intrinsic K; exemption 0d | 0.03 ± 0.01 (exempt) | 1.99 ± 0.00 (exempt) |
| discovery.fit | discovery.assignment | full fixed-K A step L32, cached evidence patterns; $K^2$ per assignment | 1.57 ± 0.08 (≤ cubic) | 0.95 ± 0.03 (measured) |
| discovery.fit | discovery.fit_family | complete two-schedule fit family L32; fresh fit memos; default 50 iterations; exemption 0c | 1.28 ± 0.08 (exempt) | 0.91 ± 0.05 (exempt) |
| discovery.search | discovery.cavity | held-out cavity L32/default 100 iterations; search budget exemption 0c remains analytic | 2.58 ± 0.30 (exempt) | 0.15 ± 0.12 (exempt) |
| discovery.release | discovery.release | fixed-assignment founder release K×32 with N carriers | 0.13 ± 0.01 (≤ cubic) | 1.03 ± 0.03 (measured) |
| selection.fit | selection.exact_pass | EXACT forward/backward pass L32, memo-free; per-pass K2 | 1.62 ± 0.06 (≤ cubic) | 0.99 ± 0.01 (measured) |
| selection.fit | selection.fit_fixed | full fit L32 with unchanged default backstop; exemption 0c; adaptive path | 1.64 ± 0.10 (exempt) | 0.96 ± 0.01 (exempt) |
| selection.search | selection.context_matching | $K^2$ context matching kernels of K×K costs per round; exemption 0; no intrinsic N | 4.01 ± 0.05 (exempt) | 0.01 ± 0.01 (exempt) |
| assembly.completion | assembly.cavity_fill | whole-bin cavity fill L64, ten-percent unknown calls, default budgets | 1.62 ± 0.13 (≤ cubic) | 0.96 ± 0.02 (measured) |
| assembly.completion | assembly.crossfit | two-fold crossfit L32, ten-percent unknown calls, default budgets | 1.34 ± 0.08 (≤ cubic) | 0.81 ± 0.03 (measured) |
| assembly.site_model | assembly.site_posterior | weak informative site, all K uncertain; exact/core route recorded | 3.06 ± 0.11 (≤ cubic) | 1.00 ± 0.00 (measured) |
| assembly.linker | assembly.linker_scan | forward micro-HMM L32 with nonuniform priors; no zero-prior memo reuse | 1.90 ± 0.03 (≤ cubic) | 0.98 ± 0.02 (measured) |
| assembly.joins | assembly.join_score | exact panel score, four blocks of four bins; structural-query count remains unresolved | 1.75 ± 0.06 (≤ cubic) | 1.02 ± 0.02 (measured) |
| assembly.shadow_refits | assembly.count_rule | K count decisions with default bootstrap; exemption 0b ranked refits separate | 1.00 ± 0.01 (exempt) | 1.28 ± 0.08 (exempt) |
| assembly.orchestration_release | assembly.release | full release of four-block K-row component without alternatives; no intrinsic N | 0.10 ± 0.01 (≤ cubic) | -0.01 ± 0.04 (measured) |
| refinement.search | refinement.score | direct exact whole-site score L64; canonical-fixing complexity remains unresolved | 1.71 ± 0.06 (≤ cubic) | 1.00 ± 0.00 (measured) |
| refinement.count | refinement.count_bound | exact one-deletion upper bounds in four blocks with K+1 rows; exemption 0b refits separate | 1.85 ± 0.12 (exempt) | 1.00 ± 0.00 (exempt) |

| Stage | Probe | What the probe covers | K slope (K = 5–30, N = 200) | N slope (N = 250–2,500, K = 15) |
|---|---|---|---|---|
| painting.coalescence | painting.transposition_gains | all K(K-1)/2 transposition gains at two links; worst candidate rechecks remain analytic | 2.39 ± 0.05 (≤ cubic) | 1.00 ± 0.00 (measured) |
| painting | painting.chain | structured sum-product, N×8 bins, Sprime=2K+1, masses enabled | 1.94 ± 0.01 (≤ cubic) | 1.00 ± 0.00 (measured) |
| painting | painting.decode_masks | both Viterbi conventions plus interval/canonical masks, N×8 bins | 1.73 ± 0.02 (≤ cubic) | 1.02 ± 0.00 (measured) |
| founder_refit | refit.alleles | conditional refit N×32, default three sweeps; random mismatches, not adversarial tie classes | 0.88 ± 0.08 (≤ cubic) | 0.71 ± 0.03 (measured) |
| pedigree.prepare | pedigree.sources | source factors and marginals, N×8 bins×$S^2$ | 1.79 ± 0.06 (≤ cubic) | 1.07 ± 0.03 (measured) |
| pedigree.screen_pair_bound | pedigree.pair_bound | all pairs of N-1 candidate parents for one child, two bins, G=S | 2.17 ± 0.16 (≤ cubic) | 2.00 ± 0.00 (measured) |
| pedigree.scores_exposure | pedigree.exposure | all N(N-1)/2 edges plus N trios, two bins, all S(S+1)/2 features | 1.58 ± 0.05 (≤ cubic) | 2.00 ± 0.00 (measured) |
| pedigree.decision | pedigree.family_messages | dense $N^2$ M0/M1 alternative family log-odds pass; iterative fit separate; no intrinsic K | 0.00 ± 0.01 (≤ cubic) | 2.01 ± 0.02 (measured) |
| pedigree.resampling_release | pedigree.resample_counts | full configured bootstrap replicate accumulator, fit kernel separate; no intrinsic K | 0.01 ± 0.01 (≤ cubic) | 1.71 ± 0.31 (measured) |
| phase.solve | phase.viterbi | all N root-sample blocks, eight bins, S=K+1; descendant paths separate | 1.93 ± 0.04 (≤ cubic) | 1.01 ± 0.00 (measured) |
| phase.release | phase.posterior | all N conditional sum-product blocks, eight bins, two gauges | 2.81 ± 0.02 (≤ cubic) | 1.00 ± 0.00 (measured) |
| recombination | recombination.follow | N meiosis chain following, 256 bins; no intrinsic K | -0.03 ± 0.02 (≤ cubic) | 1.00 ± 0.00 (measured) |
| sites_export | outputs.samples | actual BCF sample writer N×256; no intrinsic K | 0.01 ± 0.01 (≤ cubic) | 0.90 ± 0.03 (measured) |
| reports | outputs.painting_report | actual founder support tables; all K pooled rows; N×K×256 | 0.90 ± 0.02 (≤ cubic) | 0.97 ± 0.00 (measured) |
| simulation | simulation.reads | independent-site read generator N×256; K founder source only | 0.07 ± 0.01 (≤ cubic) | 0.99 ± 0.01 (measured) |
| evaluation | evaluation.founders | full founder evaluator K rows,F=2K,L=256; no intrinsic N | 1.24 ± 0.11 (≤ cubic) | -0.00 ± 0.01 (measured) |
| contracts | contracts.partners | partner changes B=32; no intrinsic N | 0.86 ± 0.07 (≤ cubic) | 0.01 ± 0.01 (measured) |
| core.ties_numerics | core.ties | workload proxy length=N*K; not a scientific K gate | 1.07 ± 0.03 (size proxy) | 1.10 ± 0.00 (size proxy) |
| kernel_warmup | core.kernel_manifest | signature encode/decode proxy count=N*K; cold compilation separate | 1.00 ± 0.00 (size proxy) | 1.00 ± 0.00 (size proxy) |
| store | store.codec | payload encode/decode proxy [N,K,32]; no scientific K gate | 0.90 ± 0.03 (size proxy) | 1.03 ± 0.02 (size proxy) |

| Stage | Probe | What the probe covers | K slope (K = 5–30, N = 200) | N slope (N = 250–2,500, K = 15) |
|---|---|---|---|---|
| engine_graph | engine.graph | independent-task validation proxy count=N*K; not scheduler throughput | 1.98 ± 0.01 (size proxy) | 1.99 ± 0.00 (size proxy) |
| interface | interface.settings | settings resolution proxy: TOML load, merge, typed build, validation and settings_text of a simulation configuration with cohorts (K, N) negligible; no intrinsic N or K | 0.00 ± 0.00 (size proxy) | -0.00 ± 0.01 (size proxy) |

**Supplementary Table S2.** The numerical ledger of release 4.0: declared normalisations in which one arithmetic form replaces several, or a declared order replaces one chosen by workers, memory, object identity or the environment. No entry changes a model, objective, threshold, candidate set, search order or budget. Source: docs/developer/ties.md.

| Entry | Category | Form | Where it acts |
|---|---|---|---|
| L1 | Formula unification | One Haldane recombination fraction, computed with `expm1`, for every map distance | genetic map, phase edge rates, pedigree selector switches |
| L2 | Summation order | One declared order (sample, natural or C order) for sums whose order otherwise followed worker counts, memory-sized batches or array memory order; the painting estimator's parameters are compared on their 2^-6 log grid | painting estimator, phase release, phase storage routes, pedigree inputs, genome likelihood, junction schedules |
| L3 | Tie order | A declared, stable content order decides exchanges between exact duplicates | assembly ties, refinement ties |
| L4 | Dtype at a test | Family phase's chain test uses float64; float32 only for storage | phase posterior and release |
| L5 | Deterministic replacement of a memory-dependent choice | The shadow-domain proof is attempted by a dimension rule, never by available memory | shadow-envelope proof, refinement count |
| L6 | Label numbering | Family numbers are canonical dense ranks ordered by each family's smallest member | relatedness release |
| L7 | Deterministic replacement of an object-identity choice | A stall's second count step runs only when the extension pool's content differs; `source_decision` names follow the decision sequence | join records, count steps |
| L8 | Environment flags | Tie reversal and the census are explicit settings passed through typed boundaries | tie settings, pedigree ties, refinement |
| L9 | Deterministic replacement of a scheduling-dependent choice | The window-work witness is the first exhausting template in the count pass's declared order | refinement work account |
| L10 | Formula unification | Founder refit qualifies and re-checks two-row moves with one direct two-row gain | refit ascent |
| L11 | Formula unification | The selection context refit's inter-bin distance is computed through the map's interval distance | context refit |

| Entry | Category | Form | Where it acts |
|---|---|---|---|
| L12 | Formula unification | Pedigree emissions use `pow(x / max, η)`, skipping the power at η = 1, on every route | child emissions |
| L13 | Formula unification | Relatedness predicts and rescores a family's log-likelihood with one function of the member sum | relatedness family terms |
| L14 | Formula unification | The template linker is compiled without fast-math reassociation; it acts only when templates are rebuilt | template linker |
| L15 | Output representation | Unphased sample genotypes are written in ascending allele order, missing first, instead of 3.5's internal order | sample phase VCF |

## Supplementary references


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